

The Effect of COVID-19 on Postsecondary Enrollment and Persistence: Evidence from North Carolina

Nir Eilam¹ Julie Edmunds² Dora Gicheva¹ Nina Arshavsky² Bryan Hutchins² Marieke Visser²

Abstract

This study leverages comprehensive administrative data on North Carolina high school graduates to analyze postsecondary enrollment and persistence patterns at 2-year and 4-year institutions before and after the onset of the COVID-19 pandemic. Tracking students for two years after their high school graduation, we observe significant shifts in these outcomes. We find a significant decline in the likelihood of enrollment at 2-year institutions during the pandemic's onset, with only partial recovery to pre-pandemic levels in subsequent years, while the likelihood of enrollment at 4-year institutions increased. Subgroup analyses indicate that Black, Hispanic, and economically disadvantaged high school graduates shifted from 2-year to 4-year institutions. We also find declines in persistence in both sectors, with sharper relative declines observed at 2-year institutions and among several student subgroups. Our regression analysis isolates deviations from pre-existing trends, suggesting that observed shifts in outcomes are not merely continuations of prior trajectories but are partly attributable to pandemic-related disruptions. This study contributes to the growing body of literature on COVID-19's impact on postsecondary education, offering the first comprehensive analysis of enrollment and persistence trends in North Carolina.

¹ Department of Economics, University of North Carolina at Greensboro

² Early College Research Center, SERVE, University of North Carolina at Greensboro

Corresponding author: Nir Eilam – n_eilam@uncg.edu

Introduction

The COVID-19 (hereafter referred to as COVID) pandemic had a profoundly disruptive impact on all aspects of life including in education. Starting in the spring of 2020 and continuing through much of the 2020-21 academic year, secondary and postsecondary institutions across the United States experienced widespread closures and rapid shifts to online teaching (Jack et al., 2023). These disruptions coincided with significant uncertainty about health, family finances, and the labor market—factors that likely shaped students’ decisions about whether and where to pursue postsecondary education. At the same time, many high schools, colleges, and universities adjusted their academic and admissions policies—modifying grading standards, adopting test-optional practices, and relaxing certain requirements—potentially lowering barriers to entry and expanding access for some students (Lovell & Mallinson, 2021; Kye & Wu, 2025). These changes likely contributed to shifts in students’ enrollment patterns and longer-term educational trajectories.

Most existing research on the effects of COVID on postsecondary education has focused on the impact on students’ immediate enrollment in postsecondary education, indicating an overall decline, with a particularly significant reduction in immediate enrollment for 2-year colleges (Bulman & Fairlie, 2022; Causey et al., 2021; Howell et al., 2021). Yet, less is known about how these disruptions influenced students’ subsequent postsecondary trajectories—including persistence and movement between institutions—and whether these effects differed across demographic and socioeconomic subgroups. Moreover, prior work has paid limited attention to distinguishing the effects of the pandemic from the continuation of pre-existing trends. This study addresses these gaps through two key research questions:

1. How did the COVID pandemic affect immediate enrollment, persistence, and transitions between 2-year and 4-year institutions, relative to pre-COVID trends?
2. How did these effects vary by student characteristics?

To address these questions, we leverage two comprehensive administrative datasets: (1) data on the universe of high school students in North Carolina from the North Carolina Department of Public Instruction, and (2) data on postsecondary enrollment from the National Student Clearinghouse. We restrict our analysis to high school graduates and construct a longitudinal dataset that tracks each graduate from 2016 to 2022, capturing their initial postsecondary enrollment and their enrollment status for the following three semesters.

We focus on three key outcomes: (1) immediate postsecondary enrollment, (2) persistence—defined as the number of semesters a student remains actively enrolled during the first three semesters after initial enrollment, and (3) postsecondary pathways, which capture how students move between different enrollment states over time. Together, these outcomes provide a comprehensive view of students' early postsecondary trajectories. We estimate the effect of COVID on these outcomes for the overall student population and across subgroups, including race, gender, and economic status, providing a nuanced view of how different populations were affected.

In our empirical strategy, we employ two regression-based approaches to analyze the first two outcomes—immediate enrollment and persistence. First, we use an event-study design to estimate changes in each outcome relative to the pre-COVID baseline year. This approach enables us to observe enrollment and persistence trends both leading up to and following the onset of the pandemic. Second, we estimate deviations from the pre-COVID trends by first modeling the trajectory of each outcome using only the pre-pandemic years. We then extend this trend forward to the COVID years, allowing us to identify breaks from the expected trajectory. While we acknowledge that these deviations may not be solely attributed to COVID, the analysis provides strong evidence of pandemic-induced disruptions.

Furthermore, the richness of our dataset enables us to control for a comprehensive set of student and high school characteristics. This allows for more precise estimation by accounting for any changes in the demographic and academic composition of graduating cohorts, ensuring that our comparisons across years reflect genuine changes in enrollment and persistence rather than underlying shifts in student populations.

Beginning with the enrollment results, we observe that the likelihood of enrollment in 4-year institutions was declining leading up to 2019 but stabilized during the COVID years, even slightly increasing. This pattern suggests that COVID may have contributed to a rise in the likelihood of enrollment in 4-year institutions. Notably, we find that the likelihood of enrollment for 2022 high school graduates was 2.2 percentage points higher than what pre-COVID trends would have predicted. For 2-year institutions, we observe that the likelihood of enrollment in these institutions was modestly declining leading up to 2019, followed by a sharp drop after the onset of COVID. This suggests that the pandemic significantly reduced the likelihood of enrollment to these institutions. Specifically, the likelihood of enrollment of 2020 high school graduates to 2-year institutions was 2.2 percentage points lower than projected based on pre-COVID trends. This decline intensified further in 2021, and showed a recovery in 2022, although not to its pre-COVID trend.

Analyzing the effect of COVID on enrollment by student and high school characteristics, we find disparate effects by subgroup. Notably, we estimate that COVID reduced the likelihood of enrollment in 2-year institutions for Black, Hispanic, American Indian, and Pacific Island and multi-racial high school graduates, relative to White high school graduates, while increasing the likelihood of Black and Hispanic high school graduates to enroll in 4-year institutions, relative to White high school graduates. This suggests that these subgroups substituted 2-year enrollment with 4-year enrollment, potentially reflecting shifting preferences or changes in access and opportunity.

Continuing with the persistence results, we find that the number of semesters that students persisted following their initial enrollment declined similarly at both 4- and 2-year institutions at the onset of the pandemic. However, due to already lower average persistence at 2-year institutions, the relative decline (compared to each sector's pre-COVID persistence level) was more pronounced in this sector. Specifically, the number of semesters persisted by students who enrolled in Fall 2019 and affected by the pandemic in their second year of studies, declined by 4.2 percent. And, whereas persistence among students at 4-year institutions returned to pre-COVID levels by Fall 2021, persistence at 2-year institutions remained well below its pre-pandemic trend. Black, Hispanic, and female students were particularly affected.

Lastly, we supplement the persistence analysis by examining the postsecondary pathways of students in each high school graduating cohort, with the results visualized in Sankey diagrams. Our findings reveal large movements between 4-year institutions, 2-year institutions, and non-enrollment, some of which have been intensified by COVID. Specifically, we observe a notable increase in the number of students from the Fall 2019 cohort who paused or dropped out after their first year, as well as a moderate rise in the proportion of students transitioning from 2-year to 4-year institutions following the onset of the pandemic.

Literature Review

Postsecondary enrollment tends to fluctuate over time, driven by external factors such as the economy and community expectations (Dunbar et al., 2011). Additionally, enrollment trends differ by sector with different patterns for 4-year and 2-year institutions. Over the period from 2012 to 2019 across the United States, the proportion of high school completers immediately enrolling in 4-year institutions had increased from 37 percent to 44 percent, with most of the jump occurring from 2012-2014 and the numbers remaining generally stable since then. In contrast, enrollment in 2-year institutions declined from 29 percent in 2012 to 22 percent in 2019 with some fluctuation in the decline (National Center for Education Statistics, 2024). Enrollment trends can also differ by subgroup (Espinosa et al., 2019). The

changing enrollment patterns over time suggest that any analyses of the impact of COVID need to account for previous enrollment patterns, especially when looking at 2-year institutions.

However, looking at students' immediate enrollment represents only a portion of students' postsecondary trajectories and can mask additional tremendous variation in students' experiences. Researchers have long acknowledged that students go in and out of postsecondary institutions (Adelman, 2006; Elliott & Hyatt, 2024); for example, almost 20 years ago, Adelman (2006) reported that a full 60 percent of undergraduates attended more than one institution with some students transferring back and forth, a phenomenon described as "swirl" (de los Santos & Wright, 1990). Researchers on this issue note that "students progress through college in unexpected patterns that include vertical, lateral, and reverse transfer, double dipping or co-enrollment, and swirl, among other patterns" (Elliott & Hyatt, 2024). There is some suggestion that transitions, and swirling may have increased during the pandemic (Fishman and Hiler, 2020), indicating that any research should examine not only immediate enrollment but students' patterns of enrollment over time, including how they transition between different educational institutions or leave postsecondary education and re-enroll.

Potential Impact of COVID on Postsecondary Enrollment

As noted earlier, the global COVID pandemic disrupted all aspects of life, including education as both secondary and postsecondary schools temporarily closed and then moved to online instruction. It is highly likely that the COVID pandemic impacted students' postsecondary decisions.

For example, as schools shifted to online learning, students may have decided that they did not want to attend college with remote learning, particularly if they were enrolling in programs with a strong hands-on component that would have been difficult to complete during the pandemic (Bulman & Fairlie, 2022). Alternatively, the online environment may have expanded access for some students, such as students with disabilities or students who might otherwise not want to go in person to a campus. The online nature may have also caused some students to switch the type of school they wanted to attend. For example, some students may have sought out community college programs instead of 4-year out of a desire to pay less for what may turn out to be similar online courses.

The impact of COVID on students' personal lives may also have affected their decision to enroll in postsecondary education. For example, many individuals experienced economic losses, especially those who started the pandemic with lower wages (Chetty et al., 2024), and students may have needed to go to work to support their families or may have had insufficient funds to enroll or persist in postsecondary education (Aucejo et al., 2020). In addition, numerous families experienced health challenges, including deaths of close relatives, that may have affected students' postsecondary enrollment decisions (Aucejo

et al., 2020), perhaps by preventing students from enrolling or encouraging them to choose a school closer to home. The stress of the pandemic also increased the mental health challenges of some students, which was associated with declines in postsecondary education plans and enrollment (Sandner et al., 2023). Lastly, the strong post-pandemic labor market could reduce community college enrollment, as some graduates may have chosen to enter the workforce rather than enroll, given abundant job openings and rapid wage growth (Johnson & Swanson, 2024).

All of this suggests that the potential impacts of COVID on postsecondary enrollment are likely to be negative, although they may not be for all populations.

Research on the Impact of COVID on Postsecondary Enrollment

Recent analyses have explored the relationship between the pandemic and postsecondary enrollment. Overall, these analyses have found that the pandemic significantly disrupted students' postsecondary plans (Liu, 2021) and that there was an "unprecedented one-year decline" in immediate postsecondary enrollment (Causey et al., 2021) with an estimated 120,000 fewer students enrolling in college immediately after high school (Howell et al., 2021). However, the impacts on postsecondary enrollment patterns differed by institutional level, and by student subgroup. Early research in the pandemic suggested that students were less likely to transfer institutions during the fall of 2020 (Causey et al., 2021) although we found no publications to date on the impact on students' persistence and transferring after that time period.

By Institutional Level

Research has generally found that immediate enrollment in community colleges dropped precipitously during the COVID period while enrollment in 4-year institutions either slightly declined or remained stable (Howell et al., 2021). An analysis of national college-level data found that enrollment of first-time students in 2-year institutions dropped 16 percent in the fall of 2020 and continued to decline in 2021; enrollment of first-time students in 4-year institutions dropped six percent and then rebounded to pre-pandemic levels by 2022 (Harris et al., 2024). A study done in California found that community college enrollment decreased in the fall of 2020 by approximately 11 percent with a drop of an additional 7 percent in the fall of 2021; however, enrollment in both the California State and University of California systems remained essentially stable during the pandemic (Bulman & Fairlie, 2022). This suggests that any analyses done on postsecondary enrollment needs to be broken out by education sector.

By Student Subgroups

Analyses completed so far suggest that certain populations were more affected by the pandemic, and that the groups most affected differed by sector. For example, Black and Latinx students had the

largest drops in community college enrollment (Bulman & Fairlie, 2022) as did first generation students (Howell et al., 2021). In contrast, drops in 4-year college enrollment rates were largest for Asian and White students and for students whose parents had a bachelor's degree (Howell et al., 2021). When looking at students' background academic performance, the drops in immediate 4-year enrollment were largest for the students with the highest GPA while the drops in 2-year enrollment were largest for students with the lowest GPA (Howell et al., 2021).

Study Contribution

This study makes several important contributions to the literature. First, to our knowledge, this is the first study to estimate the impact of COVID on the postsecondary trajectories of high school graduates in North Carolina—a large, demographically diverse state.

Second, while prior research has primarily focused on COVID's impact on enrollment alone, this study extends the analysis to include persistence and postsecondary pathways. This comprehensive approach allows us to capture the effects of COVID not only on initial enrollment but also on student retention and transitions up to two years after initial enrollment. This distinction is important, as changes in enrollment alone do not fully reflect the pandemic's impact. Indeed, we find substantial declines in student persistence due to COVID-19, revealing a more pronounced effect than what enrollment figures alone would suggest.

Third, our analysis estimates deviations from pre-COVID trends to isolate the pandemic's impact on the outcomes of interest. Specifically, we measure the extent to which postsecondary enrollment and persistence diverged from established trends after the onset of COVID. This approach accounts for the pre-existing decline in enrollment and persistence, which previous studies may have overlooked and which may have led to previous overestimates of COVID's negative impact, as declines that were a continuation of prior trends were regarded as the result of COVID. By distinguishing between pandemic-induced disruptions and ongoing trends, our findings provide a clearer assessment of the true effect of COVID.

Lastly, leveraging rich merged administrative data, we explore how the impact of COVID varied across different student demographics and high school characteristics. This nuanced analysis uncovers disparities in postsecondary trajectories that would be masked in aggregate analyses, offering important insights into which populations were most affected.

Overall, this study not only broadens the understanding of COVID's impact on postsecondary trajectories but also improves estimation accuracy by accounting for pre-existing trends and exploring differential effects across student subgroups.

Methods

Data Sources and Sample

We draw on two comprehensive administrative datasets to conduct our analysis – data on the universe of high school students in North Carolina for the school years 2015/2016 – 2021/2022 from the North Carolina Department of Public Instruction (NCDPI), and data on the universe of students in postsecondary institutions in the US for the academic years 2016/2017 – 2022/2023 from the National Student Clearinghouse (NSC). The NCDPI data include for each high school student information on their gender, race, whether they graduated from high school and if so, graduation date, whether they were classified as economically disadvantaged, disabled, and English learners during their graduation year, their graduation GPA, as well as information on their graduation high school, specifically whether it was classified as urban or not. The NSC data provides enrollment information in each semester for each student in any postsecondary institution across the US.

Starting with the NCDPI data, we limited our sample to students who graduated from high school. Then, we merged the NSC data into the NCDPI data to create a dataset that has both high school and any associated postsecondary institution data for each high school graduate in NC.¹ Our full sample includes a total of 723,517 students.

We looked at three primary outcomes: 1) immediate enrollment; 2) persistence in postsecondary education; and 3) postsecondary pathways. To look at immediate enrollment, we calculated the number of months between their high school graduation date and the date of their first enrollment into a postsecondary institution. Then, we created three dummy variables for each high school graduate - a dummy variable that takes the value 1 if the high school graduate enrolled into *any* postsecondary institution, a *4-year* postsecondary institution, and a *2-year* postsecondary institution within one year of high school graduation, and 0 otherwise.² These dummy variables are the first set of outcome variables in our regression analyses.

For the persistence analysis, we limited our sample to high school graduates who initially enrolled in postsecondary institutions in the Fall. We tracked these students for 3 semesters after their initial enrollment semester (for a total of 4 semesters), counting the number of semesters they were enrolled

¹ Approximately 2% of high school graduates from the NCDPI data did not match with high school graduates from the NSC data, so are omitted from the analysis.

² The results using 6-months rather than one year are qualitatively similar.

in any postsecondary institution.³ We did not include summer enrollment. The last cohort of enrollees that we track are those enrolled in Fall 2021 (which we track until Spring 2023). A student following a traditional path would be enrolled in 4 out of the 4 semesters for which we track them. Those enrolled in fewer than 4 four semesters either paused their education or dropped out permanently.

Lastly, to analyze postsecondary pathways, we calculated the share of each high school graduating cohort enrolled in 4-year institutions, 2-year institutions, or not enrolled in any institution in each of the 4 semesters following their high school graduation. Additionally, we calculated the magnitude of the transitions between these three enrollment statuses across these semesters (e.g., the magnitude of students who switched from 4-year institutions to 2-year institutions after their initial semester).

Analysis

We are interested in estimating the effect of COVID on enrollment and persistence outcomes. To do so, we estimate two types of regressions. First, Specification 1:

$$Y_{iht} = \sum_{\substack{t=2016 \\ t \neq 2019 \text{ or } 2018}}^{2022} \beta_t \text{Year}_t \cdot \text{Characteristic}_{ih} + \eta X_{ih} + \gamma_h + \delta_t + \varepsilon_{iht} \quad (1)$$

Where Y_{iht} is the outcome of interest – either a binary variable that takes the value 1 if high school student i graduating in year t from high school h enrolled in a postsecondary institution within 1 year of high school graduation, and 0 otherwise, or, the number of semesters that the student persisted in postsecondary institutions following their initial enrollment. We estimate this specification for three types of postsecondary institutions – any institution, 4-year institution, and 2-year institution. We include high school fixed effects, γ_h , to control for time-invariant differences between high schools that affect postsecondary enrollment (or persistence), year fixed effects, δ_t , to control for time trends, and a vector of individual controls, X_{ih} , which include a dummy variable for female, a dummy variable for each race – Black, Asian, Hispanic, American Indian, Pacific Islander, and Multi-racial (White is the reference category), a dummy variable for high school graduation within-cohort GPA quartile (highest GPA quartile is the reference category), dummy variables for economically disadvantaged, disabled, and an English learner in their graduation year, and a dummy variable for an urban high school. $\text{Year}_t \cdot \text{Characteristic}_{ih}$ are year fixed effects interacted with one of these controls. We estimate this specification separately for each of the controls (i.e. estimate separate regressions for gender, for race, etc.). Given that for the enrollment analyses we omit the year before COVID, 2019, and the reference category of each control, the

³ The count includes enrollment in any postsecondary institution, not just students' initial enrollment institutions. Thus, we allow students to switch institutions. We only count active enrollment (i.e. we do count withdrawals, etc.).

coefficients of interest, β_t 's, measure the differential effect on the likelihood of enrollment for the control, relative to the reference category, in year t , relative to the reference year, 2019. For the persistence analyses, we omit the year 2018 instead, so the coefficients of interest measure the differential effect on persisted semesters for the control, relative to the reference category, in year t , relative to the reference year, 2018.⁴ The standard errors are clustered at the high-school level.

Second, Specification 2 – referred to as the “detrended” specification – in which we proceed in two steps. First, we regress the outcome variable (i.e., one of the enrollment dummies or the persistence measure) on high school graduation year, restricting the sample to pre-COVID cohorts (2016-2019) for each subgroup of interest (e.g. in examining gender trends, we repeat this for males and females separately.). This establishes a distinct pre-COVID trend for each subgroup. We preserve the residuals from these regressions. Then, we estimate equation (1) with these residuals as the outcome variable, with each subgroup (e.g. male and female) assigned the residuals from its respective first-step regression. We then report the coefficients for the COVID years (2020-2022). This procedure essentially “deducts” the pre-COVID trend from the COVID years’ outcomes. Thus, the coefficients of interest represent deviations in outcomes during the COVID years relative to their established pre-COVID trend.

To illustrate the idea, consider an example where enrollment rates were declining leading to 2019. If this decline continued at the same rate from 2020 onward, Specification 1 would produce negative coefficients for the post-2019 years, reflecting the drop in enrollment rates relative to the 2019 baseline. In contrast, Specification 2 would yield coefficients close to zero for the same period, as the observed decline merely aligns with the established pre-COVID trend—indicating no significant deviation. We conduct this analysis to determine whether the patterns observed in the COVID years represent mere continuations of pre-existing trends—suggesting they are unlikely to be driven by the pandemic—or whether they signify breaks from pre-COVID patterns, implying that COVID may have played a contributing role.

Results

We begin by presenting the descriptive statistics, followed by the results from the two regression analyses.

Descriptive Statistics

Table 1 provides descriptive statistics for all high school graduates (Panel A) and all students enrolled in postsecondary institutions within 1 year of their high school graduation (Panel B). The statistics

⁴ The Fall 2018 enrollees are tracked through the Spring 2019 semester, which started before COVID.

are broken down by high school graduation cohort (pre-COVID – 2016-2019, and COVID– 2020-2022), and institution type. We note that any differences between statistics in the pre-COVID and COVID cohorts could reflect trends that existed prior to COVID and are, thus, not necessarily reflective of the effect of COVID. The regression analyses discussed below address this.

As Panel A reveals, the share of high school graduates that enrolled in any institution within 1 year of high school graduation decreased from 62.6% in the pre-COVID cohort to 58.6% during the COVID cohort. Breaking it down by institution reveals a larger relative drop in the share enrolled in 2-year institutions (from 25.9% to 23.2%, a 10.4% decline), compared to the share enrolled in 4-year institutions (from 36.7% to 35.4%, a 3.5% decline).

In Panel B, we examine the composition of students enrolled. We note a decline between the pre-COVID and COVID cohorts in the share of students enrolled in any institution that are White, driven mostly by a decline in 4-year institutions; a decline in the share of students that are Black, driven mostly by a decline in 2-year institutions; a decline in the share of students that are economically disadvantaged, driven mostly by a decline in 2-year institutions; an increase in the share of students that are Hispanic, driven mostly by an increase in 2-year institutions; and an increase in the share of students that were in the 2nd and 3rd quartile of their high school graduation cohort GPA, driven by increases in both 2-year and 4-year institutions. Lastly, we note that the share of enrollees that persist for 4 semesters decreased from 68.1% to 65.8%, mostly driven by 4-year institutions.

Figure 1 details the share of high school graduates that were enrolled in any postsecondary institution within 6 months of high school graduation (in blue) and one year of high school graduation (in red), by high school graduation year.⁵ Note that high school graduation year refers to the calendar year. So, for example, 2019 high school graduation year mostly includes high school students who graduated during the 2018/9 school year, usually in May or June. The figure reveals three facts. First, the trends in the share enrolled within 6 months and the share enrolled within one year are similar, which suggests it is sufficient for the analyses to focus on one of these measures, specifically the share enrolled within one year. Second, the share of high school students enrolled in any postsecondary institution has been declining monotonically leading up to COVID. While in 2016, the share was 64.0%, by 2019, the year before COVID, it dropped to 61.2%, an average yearly drop of almost one percentage point. Third, the figure reveals some of the likely effect of COVID on the share of high school students enrolled in any postsecondary institution within one year. The first high school graduating cohort during COVID, in 2020,

⁵ The exact numbers are detailed in Appendix Table A1.

had its share drop to 57.8%, a 3.4 percentage point drop. A year later, in 2021, the share slightly recovered to 58.9%.

Figure 2 details the share of students that persisted either 1,2,3, or 4 semesters following their initial enrollment into any postsecondary institution, by initial enrollment semester. We note that students initially enrolled in Fall 2019 were the first cohort in our sample affected by COVID since we track its students till Spring 2021. Examining the share who persisted for only one semester and those who persisted for the full four semesters, the former is between 7% - 7.3% in the pre-COVID years, and jumps to 7.7% in Fall 2019, and persists at its elevated level in Fall 2020 and Fall 2021, while the latter is 70.2% - 70.8% in the pre-COVID years and declines to 61.2% in Fall 2019, slightly recovering in Fall 2020, and almost completely recovering in Fall 2021.⁶

Immediate Enrollment

Overall Trends

We look at overall trends in immediate postsecondary enrollment for the NC high school student population by estimating specifications 1 and 2 with the enrollment dummy outcome variable, without interacting the year fixed effects with any controls. In contrast to the raw enrollment trends described above (and illustrated in Figure 1), these regression analyses allow us to control for individual-level and high school characteristics, thus cleansing out effects stemming from a change in the characteristics of NC high school graduates over the years.

The results from the estimation are detailed in Table 2 in which columns (1)-(3) detail the results from estimating Specification 1, separately for each type of institution, and columns (4)-(6) detail the results from estimating Specification 2, separately for each type of institution. The corresponding figures from estimating Specification 1 are presented in Figure 3.

Starting with Specification 1 – all institutions (column (1)), the coefficients for the pre-COVID years are positive, declining every year, and statistically significant at 1%. That means that leading up to COVID, the likelihood that a NC high school graduate enrolled in any postsecondary institution was declining. Examining the coefficients for the COVID cohorts, they are negative, and statistically significant ($p \leq .01$). That means that relative to the year before COVID, 2019, the likelihood that a NC high school graduate enrolled in any institution decreased. Specifically in 2020, the coefficient is -0.035, interpreted as a 3.5 percentage point decrease in the likelihood of enrollment relative to 2019. Given that the enrollment rate

⁶ These numbers are slightly different than those detailed in Table 1, as the latter are derived from the sample of all post-secondary institution enrollees, not just those who enrolled in the Fall semester.

in any institution in 2019 was 61.3%, this coefficient suggests a 5.7% decline in the likelihood of enrollment. In 2021, there is a further decline in the likelihood of enrollment relative to 2019 and in 2022, there is a slight recovery in the likelihood of enrollment relative to the sharper declines of 2020 and 2021. These coefficients are replicated in Figure 3, Panel A. Overall these results suggest a sharp decline in 2020, a further decline in 2021, and a slight recovery in 2022, though not to the pre-COVID levels. Although enrollment was declining leading up to COVID, the decline in 2020 is sharper than the decline suggested by the pre-COVID trend (clearly visible in Figure 3, Panel A), suggesting that the decline in 2020 is not merely a continuation of the pre-COVID trend, but rather a break from it, potentially due to COVID.

Examining column (2) reveals that although the likelihood of 4-year enrollment saw a small decline in 2020 (-0.0057***), the coefficient for 2021 is positive and not statistically significant and that of 2022 is positive and statistically significant at 5%. In contrast, the likelihood of 2-year enrollment saw a large decline in 2020 (-0.029***), suggesting a 11.2% decline in the likelihood of enrollment relative to the 2019 mean, a further decline in 2021 (-0.037***), and moderated decline in 2022 (-0.024***), all relative to 2019. These suggest that the decline in the likelihood of enrollment in any institution in the COVID years stemmed primarily from a decline in the likelihood of enrollment in 2-year institutions.

Next, we examine the results of estimating specification 2 in columns (4)-(6). Starting with column (4) – all institutions, the coefficients for 2020 and 2021 are negative and statistically significant (-0.024*** and -0.016***, respectively) but are smaller in magnitude than the corresponding coefficients from specification 1. This is due to the fact that the likelihood of enrollment in any institution was already declining leading up to 2019. If we assume that the pre-COVID trend would have continued to the COVID years, this means that part of the COVID years coefficients estimated in specification 1 were due to a pre-existing trend, and not a deviation from the trend. By 2022, the coefficient becomes positive and statistically significant (0.011***) suggesting a recovery in likelihood of enrollment relative to the trend. Examining columns (5) and (6) reveals that, similarly to specification 1, the decline in the likelihood of enrollment in any institution was driven by a decline in the likelihood of enrollment in 2-year institutions, whereas the likelihood of enrollment in 4-year institutions actually increased in 2021 and 2022 relative to what the pre-COVID trend would have suggested. Specifically, the coefficients of 0.10 and 0.22 in 2021 and 2022, respectively, suggest that the likelihood of enrollment in 4-year institutions rose by 2.8% and 6.2% compared to its 2019 mean.

Subgroup Analyses

We examine immediate enrollment trends by subgroups by estimating specifications 1 and 2 separately for each characteristic of interest (e.g. race, disability status, etc.) including the interaction of the year fixed effects with that characteristic, while omitting the reference category (e.g. Male).

We focus the analyses on the results of estimating specification 2, which estimates the deviations in the COVID years from the pre- COVID trend. These results for all institutions, 4-year institutions, and 2-year institutions, are presented in Table 3 Panels A, B, and C, respectively, with each regression separated by a dashed vertical line, and the year coefficients (representing the year effects, relative to the pre-COVID trend, for the omitted category) and the interaction coefficients (representing the year effects, relative to the pre-COVID trend, for each respective category, relative to the omitted category) are reported in the rows. The results from estimating specification 1 for all institutions, 4-year institutions, and 2-year institutions, are presented in Appendix Figures A1, A2, and A3, respectively; the corresponding coefficients are detailed in Appendix Table A2 Panels A, B, and C, respectively, with each regression separated by a dashed vertical line, and the interaction coefficients for each year and the control variable reported in the rows.

Gender (column (1)): The year coefficients indicate that male enrollment (the omitted category) in any institution declined in 2020 and 2021 and increased in 2022 relative to its pre-COVID trend. The interaction coefficients indicate that female enrollment experienced a smaller decline in 2020 (0.0076***), but larger declines in 2021 and 2022, relative to male enrollment (-0.015*** and -0.012***, respectively). These results are driven solely by 2-year institutions.

Race (column (2)): The enrollment of White high school graduates (the omitted category) in any institution declined in 2020 and 2021 relative to its pre-COVID trend and recovered back to its pre-COVID trend in 2022. Relative to White enrollment, Black enrollment increased in growing magnitudes in all COVID years (0.0068*, 0.018***, and 0.024***, in 2020, 2021, and 2022, respectively). These results were driven by large relative increases in Black enrollment to 4-year institutions (0.025***, 0.039***, and 0.033***), while relative Black enrollment to 2-year institutions declined albeit in smaller magnitudes (-0.018***, -0.021***, and -0.0094**). In contrast, relative to White enrollment, Asian enrollment decreased in 2021, and 2022 in growing magnitudes (-0.014*, and 0.040*** respectively). These results were driven by declines in Asian enrollment to 4-year institutions, while Asian enrollment to 2-year institutions increased although in smaller magnitudes. With respect to Hispanic enrollment, there was a decline in enrollment, relative to White enrollment, in 2020 (-0.010**), followed by no differences in 2021 and 2022. By sector, Hispanic enrollment increased in 4-year institutions in 2020 and 2021 and decreased in 2-year institutions in those years, relative to White enrollment. With respect to the other races, we

note decreases in American Indian and Pacific Islander enrollment in all institutions, relative to White enrollment, driven mostly by declines in enrollment to 2-year institutions. The decline of Pacific Islander enrollment in all institutions, relative to White enrollment, is particularly large in 2021 and 2022 (-0.2*** and -0.24***, respectively).

GPA (column (7)): The enrollment of high school graduates with high school GPAs in the fourth quartile (The quartile with the highest grades; the omitted category) in any institution declined in 2020 and 2021, relative to its pre-COVID trend, and recovered back to its pre-COVID trend in 2022. Relative to these students, high school graduates with GPAs in the first, second, and third quartile, experienced declines in enrollment in 2020 (-0.015***, -0.023***, and -0.015***, respectively), and an increase in enrollment in 2022 (0.021***, 0.014***, and 0.028***), both mainly driven by changes in enrollment of high school graduates in these quartiles to 2-year institutions.

Economically Disadvantaged (columns (3)): The enrollment of non-ED high school graduates (the omitted category) in any institution declined in the COVID years relative to its pre-COVID trend. While in 2020 ED enrollment declined even further, relative to non-ED enrollment (-0.01***), 2021 and 2022 ED enrollment declined less than non-ED enrollment (0.089** and -0.0078*, respectively). The enrollment of economically disadvantaged (ED) high school graduates to 4-year institutions increased in 2020, 2021 and 2022, whereas their enrollment to 2-year institutions decreased in 2020 and 2021.

Other Characteristics (columns (4)-(6)): The enrollment of non-disabled, non-English learners (EL), and non-urban students (the omitted categories) in any institution declined in 2020 and 2021, relative to its pre-COVID trend, and recovered to its pre-COVID trend in 2022. While the enrollment trends of disabled and EL students were mostly similar to non-disabled and non-EL students, the enrollment of urban students declined less than the enrollment of non-urban students in all COVID cohorts.

Persistence

Overall Trends

For the persistence analyses, we estimate specifications 1 and 2 with the number of semesters persisted in postsecondary institutions as the outcome, limiting the sample to high school graduates who initially enrolled in postsecondary institutions in the Fall. We begin by estimating overall persistence trends. The results from the estimation are detailed in Table 4 in which columns (1)-(3) detail the results from estimating specification 1, separately for each type of institution, and columns (4)-(6) detail the results from estimating specification 2, separately for each type of institution. The corresponding figures from estimating specification 1 are presented in Figure 4.

Starting with specification 1 – all institutions (column (1)), the coefficients for the pre-COVID years are positive, declining every year, and statistically significant. Meaning that leading up to COVID, the number of persisted semesters was declining. In the COVID years, the coefficients are negative for all years, and statistically significant ($p \leq .01$). That is, the number of persisted semesters continued declining during the COVID cohorts in magnitudes larger than prior declines. Specifically, in Fall 2019, the coefficient is -0.11, interpreted as a 0.11 decline in persisted semesters (out of 4), relative to Fall 2018. In Fall 2020, the coefficient declines further to -0.15 and recovers slightly to -0.062 in Fall 2021, but not to its pre-COVID level. Examining columns (2) and (3) reveals similar patterns across 2-year and 4-year institutions with slightly larger declines in persistence in Fall 2019 and 2020 and a sharper recovery in Fall 2021 in 4-year institutions. However, the coefficient for the pre-COVID years reveal that persistence was declining slightly faster leading up to COVID in 4-year institutions.

Continuing with specification 2 – all institutions (column (4)) reveals similar patterns when accounting for pre-COVID trends – a sharp decline in persistence in Fall 2019 relative to the trend, further decline in Fall 2020, and a recovery in Fall 2021. The coefficients are all statistically significant ($p \leq .01$). Specifically, in Fall 2019, the coefficient is -0.12, interpreted as a 0.12 decline in persisted semesters (out of 4), relative to the pre-COVID trend. Given that the number of persisted semesters pre-COVID across all institutions was 3.45, this suggests a 3.5% decline in persisted semesters relative to the pre-COVID trend that would have prevailed absent of COVID. By Fall 2021, persistence recovers slightly but is still below its pre-COVID trend (the coefficient for Fall 2021 is -0.074***). Examining columns (5) and (6) reveals similar patterns across 2-year and 4-year institutions in persistence in Fall 2019 and Fall 2020 with larger declines in 2-year institutions, and a sharper recovery in 4-year institutions, such that by Fall 2021, persistence in 4-year institutions is back to its pre-COVID trend (0.0022), whereas persistence in 2-year institutions is still significantly below its pre-COVID trend (-0.17***). Moreover, given the lower mean persistence in 2-year institutions compared to 4-year institutions in the pre-COVID period (3.07 and 3.68, respectively), the larger coefficients for the former also suggest an even larger relative (i.e. percent) decline in persistence.

The fact that in specification 2, the negative effect on persistence is larger for 2-year institutions compared to 4-year institutions, whereas the opposite is true in specification 1, stems from the sharper pre-COVID decreasing trend in persistence in 4-year institutions. That is, more of the decline in persistence observed in 4-year institutions is a continuation of the pre-COVID trend, rather than the potential effect of COVID.

Subgroup Analyses

We examine persistence trends by subgroups by repeating a similar analysis to that described in prior for the enrollment outcome. Similarly, we focus the analyses on the results of estimating specification 2, which estimates the deviations in the COVID years from the pre-COVID trend. The results for all institutions, 4-year institutions, and 2-year institutions, are presented in Tables 5 Panels A, B, and C, respectively, with each regression separated by a dashed vertical line, and the year coefficients (representing the year effects, relative to the pre-COVID trend, for the omitted category) and the interaction coefficients (representing the year effects, relative to the pre-COVID trend, for each respective category, relative to the omitted category) are reported in the rows.⁷

Gender (column (1)): The year coefficients indicate that the persistence of male students (the omitted category) in any institution declined in the COVID years relative to its pre-COVID trend. The interaction coefficients indicate that while Fall 2019 female persistence did not decline differentially than male persistence, Fall 2020 and 2021 female persistence experienced a steeper decline than male persistence (-0.023** and -0.064***, respectively). This decline was more pronounced in 2-year institutions than in 4-year institutions.

Race (column (2)): The persistence of White students (the omitted category) in any institution declined in the COVID years relative to its pre-COVID trend. Several other racial groups experienced even larger declines in persistence. Black persistence declined more than White persistence in each of the COVID cohorts (-0.027**, -0.073***, and -0.069***, in Fall 2019, 2020, and 2021, respectively), with larger relative declines in 2-year institutions than in 4-year institutions. Hispanic persistence also declined more than White persistence in each of the COVID cohorts (-0.047***, -0.11***, and -0.079***, in Fall 2019, 2020, and 2021, respectively), with results driven almost solely by relative declines in 4-year institutions rather than 2-year institutions. Similarly, Asian persistence declined more than White persistence in Fall 2019 and Fall 2020 (-0.039** and -0.06***, respectively), with results driven solely by relative declines in 4-year institutions rather than 2-year institutions. By Fall 2021, Asian persistence exhibited no differential decline relative to white persistence. With respect to the other races, we note increases in American Indian persistence and decreases in Multiracial persistence relative to White persistence in some, or all, COVID years and differential declines in Pacific Islander persistence relative to White persistence.

GPA (column (7)): The persistence of students with HS GPAs in the fourth quartile (The quartile with the highest grades; the omitted category) in any institution increased in Fall 2020, and decreased in Fall 2020, relative to its pre-COVID trend, and by Fall 2021, returned to its pre-COVID trend. The

⁷ The results from estimating specification 1 are available by request.

persistence trends of students with lower HS GPAs were mostly similar with the exception of Fall 2021 students with HS GPAs in the second and first quartiles whose persistence declined more than those in the fourth quartile (-0.11^{***} and -0.1^{***} , respectively) .

Economically Disadvantaged (column (3)): The persistence of non-ED students (the omitted category) in any institution declined in the COVID years relative to its pre-COVID trend. While Fall 2019 ED persistence did not decline differentially than non-ED persistence, Fall 2020 and 2021 ED persistence experienced a larger decline than non-ED persistence (-0.043^{***} and -0.036^{***} , respectively).

Other Characteristics (columns (4)-(6)): The persistence of non-disabled, non-English learners (EL), and non-urban students (the omitted categories) in any institution declined in the COVID years relative to its pre-COVID trend. While the persistence trends of EL and urban students were mostly similar to non-EL and non-urban students, the persistence of disabled students declined more than the persistence of non-disabled students in all COVID cohorts.

Postsecondary Paths

The above analysis summarizes students' persistence into a single measure, the number of persisted semesters. While this measure provides a general overview of student retention in postsecondary institutions after enrollment, it lacks insight on several specific dimensions. First, it does not provide insights on movements between postsecondary institutions. For example, a student who persists in their initial enrolled institution is not distinguished from a student who transfers from their initial enrolled institution to another institution. Second, it does not reveal insights on the timing of the movements.

To explore these dimensions, we supplement the persistence analysis with an examination of the postsecondary paths taken by students in each high school graduating cohort, visualized using Sankey diagrams, provided in Figure 5.

A separate diagram was created for each high school graduating cohort. The diagram tracks the flow of high school graduates across different enrollment states – 4-year institutions, 2-year-institutions, and “Other” (not enrolled in any postsecondary institution) – over the four semesters following their high school graduation; summer semesters were not included. For example, those who graduated in May 2019 were tracked through the Spring 2021 semester. In each semester, their enrollment status was recorded, and was used to calculate the share of students, out of the high school graduating cohort, in each state. These are the shares detailed in the diagram. Additionally, each student's enrollment status in a given semester was compared to their enrollment status in the previous semester to identify whether they

remained in the same state (e.g., staying enrolled in a 4-year institution) or transitioned to a different state (e.g. transferring from a 4-year institution to a 2-year institution). These transitions are visualized through lines between nodes, with the thickness of each line proportional to the share of students making that transition.

To illustrate this, consider the Sankey diagram for the 2018 cohort. Out of all 2018 high school graduates, 39% enrolled in 4-year institutions, 22% enrolled in 2-year institutions, and 40% did not enroll in any postsecondary institution in the semester following graduation, Fall 2018. The shares do not sum up to 100% due to rounding. In the next semester, Spring 2019, 38% of all 2018 high school graduates were enrolled in 4-year institutions, a decline of approximately 1 percentage point. Those include students who remained enrolled in 4-year institutions, represented by the incoming thick green line; students who transferred from 2-year institutions, represented by the incoming orange line, and students enrolling for the first time after not being enrolled in Fall 2018, represented by the incoming purple line.

We note that the 2018 cohort was not affected by COVID, since it was tracked until the Spring 2020; the 2019 cohort was partially affected, beginning in their second tracked semester (Spring 2020); and the 2020 and 2021 cohorts were fully affected, although COVID infections started declining for the latter part of their tracked semesters.

The diagrams of the four high school graduating cohorts reveal several insights. First, the share of high school graduates who did not enroll in any postsecondary institution immediately after graduation was highest in the 2020 cohort (43%), followed by a slight decrease in the 2021 cohort (42%). This is consistent with the Immediate Enrollment analysis above. The share of high school graduates who were not enrolled in any postsecondary institution in their fourth tracked semester was also the highest for the 2020 cohort (51%), followed by the 2019 and 2021 cohorts (50% and 49%, respectively), all higher than that share for the 2018 cohort (47%).

Second, across all cohorts, the largest movements between states are those from “Other” to 2-year institutions (i.e. high school graduates who enroll in 2-year institutions several semesters after graduation), and from 2-year institutions to “Other” (i.e. students enrolled in 2-year institutions who either drop-out or pause their studies). The latter movements are particularly large after each cohort’s first year of studies. We elaborate on these movements, as well as other movements, next.

Based on semester-by-semester enrollment data used for the Sankey diagrams, we calculated student transitions between states—4-year institutions, 2-year institutions, and “Other”—across four semesters for each graduating cohort. Table 6 summarizes these movements by the transition type: from

postsecondary institutions to “Other” (Panel A), between 2-year and 4-year institutions (Panel B), and from “Other” to postsecondary institutions (Panel C).

In Panel A, the left number is the share switching from 4-year institutions to “Other” and the right number is the share switching from 2-year institutions to “Other”. In Panel B, the left number is the share switching from 4-year institutions to 2-year institutions and the right number is the share switching from 2-year institutions to 4-year institutions. In Panel C, the left number is the share of students switching from “Other” to 4-year institutions and the right number is the share of students switching from “Other” to 2-year institutions.

These shares are calculated out of students enrolled in the originating state during the previous semester. They include both permanent and temporary transitions; for example, students who exit to “Other” and later return are still counted as having exited in the semester the transition occurred. Likewise, changes in these shares may reflect either differences in the number of movers or shifts in the composition (“stock”) of students in each enrollment state.

Panel A presents the share of students who transitioned from 4-year or 2-year institutions to the “Other” category between semesters. Students enrolled in 2-year institutions were substantially more likely to exit to “Other” than those in 4-year institutions—by a factor of up to 4.3 across cohorts. These transitions peak after the second semester (i.e., after the first year of study), particularly among students in 2-year institutions, where exit rates ranged from 25% to 34%. The cohort with the highest exit rates after their second semester, as well as after their third semester, is the Fall 2019 cohort, that was indeed affected by COVID for the first time in their second semester of studies. Among the cohorts fully affected by COVID (2020 and 2021), transitions to “Other” after the first semester were slightly elevated, but transitions to “Other” after the second and third semester were somewhat lower compared to the 2019 cohort. This may reflect a form of selection—where students most vulnerable to dropout did not enroll at all in Fall 2020 and 2021—or greater awareness of pandemic-related challenges at the time of initial enrollment.

Panel B summarizes movements between 2-year and 4-year institutions. Upward transfer from 2-year to 4-year institutions was more common than the reverse across all cohorts, with the highest switching rates occurring after the second semester. COVID-era cohorts (2020 and 2021) show a modest increase in 2-to-4-year transfers compared to pre-COVID cohorts, particularly after the second and third semesters. These patterns may suggest that students delayed transfer during the early stages of the pandemic and resumed those plans as conditions stabilized.

Panel C presents the share of students who transitioned from “Other” to 4-year and 2-year institutions between semesters. This includes students who enrolled for the first time or those who re-enrolled in postsecondary education after spending a semester or more in the “Other” category. Transitions in this direction were most frequent after students’ first and second semesters, with re-entry consistently higher into 2-year institutions compared to 4-year institutions. Comparisons across cohorts reveal no substantial differences in these patterns.

Taken together, these tables reveal the value of going beyond a single persistence metric. Transition patterns differ not only by institution type but also by semester and cohort, highlighting important differences in students’ experiences of interruption, transfer, and delayed enrollment.

Discussion and Conclusion

This study offers a nuanced understanding of how the COVID pandemic impacted postsecondary enrollment and persistence among North Carolina high school graduates. By leveraging comprehensive administrative datasets, the analysis reveals shifts in both immediate enrollment and short-term persistence patterns across 2-year and 4-year institutions. These changes are not mere continuations of pre-existing trends; rather, they reflect distinct breaks attributable to pandemic-induced disruptions.

The pandemic led to a decline in 2-year institution enrollment, with sharp drops observed for 2020 and 2021 high school graduates, followed by a partial recovery in 2022—though not to its pre-COVID trend. In contrast, 4-year institution enrollment rose modestly above its pre-COVID trend. Subgroup analyses underscore important disparities: Black, Hispanic, and economically disadvantaged students exhibited notable shifts from 2-year to 4-year institutions. These shifts may reflect several factors, including institutional changes such as test-optional policies and flexible admissions, evolving student preferences under uncertainty, and broader structural changes in access to higher education.

Persistence declined across both 2-year and 4-year institutions during the pandemic, with sharper drops in the 2-year sector. This is consistent with evidence by Gicheva et al. (2025) of persistent decreases in course completion and credit earning—particularly in 2-year institutions. This divergence highlights underlying vulnerabilities in community colleges that may have been exacerbated by the pandemic’s economic and social disruptions. Qualitative evidence by Hutchins et al. (2025) from students and advisors further indicates that many undergraduates struggled with online instruction, social isolation, and reduced academic preparation, contributing to higher withdrawal rates and disrupted degree progress. Our subgroup analyses reveal that Black, Hispanic, and economically disadvantaged

students—groups historically more reliant on community colleges as accessible entry points—were disproportionately affected, emphasizing the need for stronger supports to sustain student success beyond initial enrollment.

Lastly, the observed transitions between 4-year institutions, 2-year institutions, and non-enrollment indicate somewhat greater movement across these states during the pandemic, particularly from enrollment to non-enrollment and from 2-year to 4-year institutions. The increased transition from 2-year institutions to non-enrollment after the first year imply that some students struggled with remote instruction, financial strain, and personal hardships during this period. Conversely, the modest rise in upward transfers to 4-year institutions points to resilience and potentially strategic adaptation among students seeking better credentials amid economic uncertainty. This pattern may also reflect the relaxation of admission requirements and broader access policies implemented by some 4-year institutions, which may have expanded transfer opportunities.

This study demonstrates how COVID-19 reshaped the early postsecondary experiences of North Carolina high school graduates and highlights important differences in how various subgroups navigated these disruptions, with implications for equity and access in postsecondary education. Given that the most pronounced disruptions occurred in the community college sector, future efforts should prioritize this sector, which serves as a critical entry point for low-income and first-generation students. At the same time, the pandemic also altered pathways into 4-year institutions, and although access to 4-year institutions expanded for some groups, our findings suggest that access alone is insufficient without targeted measures to promote persistence. Policymakers and institutions should invest in advising, academic support, and financial stability programs to ensure that these enrollment gains translate into degree completion.

Future work should extend this analysis to examine outcomes beyond short-term persistence, such as degree completion and labor market entry, and explore whether the enrollment shifts we document translate into lasting changes in postsecondary inequality. As higher education continues to adjust in the wake of the pandemic, evidence-based policy design will be important for supporting both access and persistence.

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Tables

Table 1: Descriptive Statistics

	2016-2019			2020-2022		
	Any Institution	4-year	2-year	Any Institution	4-year	2-year
Panel A: High School Graduates						
Enrolled Within One Year	62.6	36.7	25.9	58.6	35.4	23.2
Observations	414,799			308,718		

Panel B: Students Enrolled in Postsecondary Institutions Within One Year of Their High School Graduation

Female	55.6	57.6	52.8	56.2	58.2	53.0
White	57.6	59.7	54.6	56.3	57.9	53.8
Black	22.6	23.4	21.4	20.8	22.1	18.8
Hispanic	9.9	6.3	14.9	12.9	8.9	19.0
Asian	3.6	4.4	2.5	4.5	5.6	2.7
American Indian	0.96	0.79	1.2	0.85	0.75	0.99
Pacific Islander	0.05	0.04	0.06	0.08	0.08	0.07
Multiracial	5.3	5.3	5.4	4.6	4.7	4.6
Economically Dis.	28.1	22.2	36.5	20.1	17.0	26.2
Disabled	4.7	2.0	8.5	4.6	2.1	8.4
English Learners	1.1	0.24	2.3	2.1	0.71	4.2
Urban	39.5	43.3	34.2	41.5	44.7	36.7
GPA Quartile 1	10.5	2.8	21.4	9.7	2.5	20.7
GPA Quartile 2	19.9	12.5	30.6	21.1	13.9	32.1
GPA Quartile 3	27.8	27.7	28.0	30.6	30.3	31.0
Enrollees Persisting Four Semesters	68.1	78.6	51.7	65.8	74.1	51.5
Observations	259,757	152,203	107,539	180,898	109,234	71,662

Notes: Panels A and B provide descriptive statistics on all HS graduates, and all high school graduates enrolled in postsecondary education within one year of their graduation, respectively. The descriptive statistics are broken down by high school graduation cohorts (2016-2019 and 2020-2022) and by the type of initial postsecondary institution (any institution, 4-year, and 2-year). All numbers, except the number of observations, represent shares.

Table 2: Enrollment Regression Results – Overall Trends

	Specification 1			Specification 2		
	(1) All Institutions	(2) 4-year	(3) 2-year	(4) All Institutions	(5) 4-year	(6) 2-year
2016	0.027*** (0.0025)	0.021*** (0.0023)	0.0058** (0.0026)	- -	- -	- -
2017	0.018*** (0.0023)	0.020*** (0.0022)	-0.0021 (0.0022)	- -	- -	- -
2018	0.006*** (0.0022)	0.015*** (0.0021)	-0.0089*** (0.0023)	- -	- -	- -
2019 (omitted)	- -	- -	- -	- -	- -	- -
2020	-0.035*** (0.0021)	-0.0057*** (0.0020)	-0.029*** (0.0024)	-0.024*** (0.002)	-0.0026 (0.0018)	-0.022*** (0.0018)
2021	-0.037*** (0.0024)	0.00018 (0.0021)	-0.037*** (0.0026)	-0.016*** (0.0021)	0.010*** (0.0018)	-0.027*** (0.0021)
2022	-0.019*** (0.0026)	0.0049** (0.0023)	-0.024*** (0.0028)	0.011*** (0.0023)	0.022*** (0.0019)	-0.010*** (0.0024)
Mean	0.613	0.353	0.260	0.613	0.353	0.260
Observations	712,588	712,588	712,588	712,588	712,588	712,588
Adj. R-squared	0.29	0.38	0.06	0.29	0.38	0.06

Notes: This table presents the estimation results for Specification 1 (columns (1)-(3)) and Specification 2 (columns (4)-(6)). Each column corresponds to a separate regression using one of the three enrollment dummy outcome variables: any institution, 4-year institution, and 2-year institution. Interaction of year and each control is not included in the regression. Estimates are reported for each year, with 2019 serving as the reference year. The mean is calculated in 2019. Robust standard errors, clustered at the high school level, are provided in parentheses. *** p<0.01; ** p<0.05; * p<0.1.

Table 3: Enrollment Regression Results – Subgroup Analysis, Specification 2

(A) All Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
2020	-0.028*** (0.0025)			-0.028*** (0.0024)				-0.017*** (0.0022)	-0.024*** (0.0021)	-0.029*** (0.002)	-0.028*** (0.0027)		-0.011*** (0.0023)	
2021	-0.009*** (0.0027)			-0.025*** (0.0026)				-0.014*** (0.0023)	-0.018*** (0.0022)	-0.023*** (0.0022)	-0.023*** (0.0028)		-0.011*** (0.0029)	
2022	0.017*** (0.0029)			-0.0027 (0.0026)				0.015*** (0.0025)	0.010*** (0.0024)	0.0028 (0.0023)	0.0050* (0.0029)		-0.0034 (0.0028)	
2020 Interaction	0.0076*** (0.0028)	0.0068* (0.0041)	-0.0017 (0.0079)	-0.010** (0.0048)	-0.042*** (0.014)	-0.019 (0.047)	0.001 (0.0068)	-0.010*** (0.0035)	-0.012** (0.0052)	-0.011 (0.0091)	0.013*** (0.0039)	-0.015*** (0.0041)	-0.023*** (0.0044)	-0.015*** (0.0039)
2021 Interaction	-0.015*** (0.0029)	0.018*** (0.004)	-0.014* (0.0078)	-0.0081 (0.0049)	-0.046*** (0.012)	-0.20*** (0.053)	-0.0032 (0.0075)	0.0089** (0.0038)	0.007 (0.0054)	-0.0076 (0.0082)	0.021*** (0.0042)	-0.0042 (0.0045)	-0.014*** (0.0048)	-0.0013 (0.0039)
2022 Interaction	-0.012*** (0.003)	0.024*** (0.0045)	-0.040*** (0.008)	0.0085 (0.0056)	-0.078*** (0.016)	-0.24*** (0.06)	0.014* (0.0072)	0.0078* (0.0042)	-0.004 (0.0062)	-0.022*** (0.0085)	0.020*** (0.0047)	0.021*** (0.0049)	0.014*** (0.0051)	0.028*** (0.0042)
Mean	0.545			0.681				0.700	0.642	0.624	0.603		0.913	
Observations	712,588			712,588				712,588	712,588	712,588	712,588		712,588	
Adj. R-squared	0.29			0.29				0.29	0.29	0.29	0.29		0.05	

(B) 4-Year Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
2020	-0.0029 (0.0019)			-0.016*** (0.0023)				-0.0027 (0.0022)	-0.0032* (0.0019)	-0.008*** (0.0018)	-0.0047** (0.0024)		-0.004 (0.0034)	
2021	0.012*** (0.0021)			-0.0075*** (0.0024)				0.010*** (0.0021)	0.0095*** (0.0019)	0.0028 (0.0018)	0.0069*** (0.0023)		0.010*** (0.0037)	
2022	0.023*** (0.0023)			0.004 (0.0026)				0.025*** (0.0022)	0.023*** (0.002)	0.013*** (0.0019)	0.022*** (0.0023)		0.030*** (0.0037)	
2020 Interaction	0.00059 (0.0029)	0.025*** (0.0035)	-0.013* (0.0068)	0.014*** (0.0038)	-0.029** (0.014)	0.024 (0.048)	0.012* (0.0062)	0.011*** (0.0029)	-0.0014 (0.0036)	0.0019 (0.005)	0.0089*** (0.0034)	0.0084** (0.0037)	-0.0026 (0.0044)	0.0056 (0.0045)
2021 Interaction	-0.003 (0.0028)	0.039*** (0.0038)	-0.041*** (0.0088)	0.0082** (0.0038)	-0.012 (0.011)	-0.064 (0.044)	0.0096 (0.0068)	0.018*** (0.0029)	-0.004 (0.004)	0.0025 (0.0054)	0.013*** (0.0036)	0.00072 (0.0041)	-0.00088 (0.0047)	0.0087* (0.0049)
2022 Interaction	-0.0021 (0.0029)	0.033*** (0.0044)	-0.068*** (0.0098)	0.0038 (0.0043)	-0.024* (0.012)	-0.080* (0.048)	0.017** (0.0068)	0.011*** (0.0034)	-0.023*** (0.0041)	-0.018*** (0.0058)	0.0061 (0.0039)	-0.020*** (0.0042)	-0.018*** (0.0049)	0.015*** (0.0049)
Mean	0.299			0.410				0.439	0.380	0.364	0.323		0.768	
Observations	712,588			712,588				712,588	712,588	712,588	712,588		712,588	
Adj. R-squared	0.29			0.29				0.29	0.29	0.29	0.29		0.29	

(C) 2-Year Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
2020	-0.025*** (0.0024)			-0.012*** (0.0024)				-0.014*** (0.0022)	-0.020*** (0.002)	-0.021*** (0.0019)	-0.024*** (0.0023)		-0.0065** (0.0028)	
2021	-0.021*** (0.0027)			-0.017*** (0.0026)				-0.024*** (0.0023)	-0.028*** (0.0022)	-0.026*** (0.0021)	-0.030*** (0.0026)		-0.021*** (0.003)	
2022	-0.0053* (0.0029)			-0.0068** (0.0028)				-0.01*** (0.0025)	-0.012*** (0.0024)	-0.010*** (0.0024)	-0.017*** (0.0029)		-0.034*** (0.0032)	
2020 Interaction	0.0068** (0.0032)	-0.018*** (0.0037)	0.012 (0.0076)	-0.024*** (0.0047)	-0.013 (0.011)	-0.043 (0.046)	-0.011* (0.0066)	-0.021*** (0.0035)	-0.011** (0.0051)	-0.012 (0.0088)	0.004 (0.0039)	-0.023*** (0.0044)	-0.020*** (0.0044)	-0.020*** (0.0041)
2021 Interaction	-0.012*** (0.0031)	-0.021*** (0.004)	0.027*** (0.0083)	-0.016*** (0.0049)	-0.034*** (0.013)	-0.13*** (0.041)	-0.013* (0.0068)	-0.0090** (0.0036)	0.011** (0.0053)	-0.01 (0.0086)	0.0074* (0.0044)	-0.0052 (0.0047)	-0.014*** (0.0048)	-0.010** (0.0045)
2022 Interaction	-0.010*** (0.0032)	-0.0094** (0.0044)	0.027*** (0.0093)	0.0048 (0.0055)	-0.054*** (0.019)	-0.16*** (0.044)	-0.0033 (0.007)	-0.0037 (0.0039)	0.019*** (0.0058)	-0.004 (0.0088)	0.013*** (0.0048)	0.040*** (0.0049)	0.032*** (0.0056)	0.013*** (0.0044)
Mean	0.246			0.270				0.261	0.262	0.260	0.279		0.145	
Observations	712,588			712,588				712,588	712,588	712,588	712,588		712,588	
Adj. R-squared	0.29			0.29				0.29	0.29	0.29	0.29		0.29	

Notes: These tables present the estimation results for Specification 2 using one of the enrollment dummy outcomes variables: any institution (Panel A), 4-year institution (Panel B), and 2-year institution (Panel C). Each regression is separated by a dashed vertical line. Estimates are reported for each year and the interactions of year and the control variable. The mean is calculated in 2019 for the omitted category (e.g. male for column (1)). Robust standard errors, clustered at the high school level, are provided in parentheses. *** p<0.01; ** p<0.05; * p<0.1.

Table 4: Persistence Regression Results – Overall Trends

	Specification 1			Specification 2		
	(1) All Institutions	(2) 4-year	(3) 2-year	(4) All Institutions	(5) 4-year	(6) 2-year
Fall 2016	0.029*** (0.0058)	0.036*** (0.0056)	0.025** (0.0012)	- -	- -	- -
Fall 2017	0.012** (0.0053)	0.023** (0.0056)	-0.0039 (0.011)	- -	- -	- -
Fall 2018 (omitted)	- -	- -	- -	- -	- -	- -
Fall 2019	-0.11*** (0.0055)	-0.12*** (0.006)	-0.095*** (0.01)	-0.12*** (0.0049)	-0.11*** (0.0054)	-0.13*** (0.0082)
Fall 2020	-0.15*** (0.006)	-0.16*** (0.007)	-0.14*** (0.01)	-0.16*** (0.0051)	-0.13*** (0.0058)	-0.18*** (0.0088)
Fall 2021	-0.062*** (0.005)	-0.040*** (0.006)	-0.11*** (0.012)	-0.074*** (0.0051)	0.0022 (0.0051)	-0.17*** (0.01)
Mean	3.45	3.68	3.07	3.45	3.68	3.07
Observations	345,748	213,248	132,478	345,748	213,248	132,478
Adj. R-squared	0.25	0.24	0.14	0.25	0.24	0.14

Notes: This table presents the estimation results for Specification 1 (columns (1)-(3)) and Specification 2 (columns (4)-(6)) using the persistence outcome variable. Each column corresponds to a separate regression for each type of institution: any institution, 4-year institution, and 2-year institution. Interaction of year and each control is not included in the regression. Estimates are reported for each enrollment cohort, with Fall 2018 serving as the reference enrollment cohort. The mean is calculated in Fall 2018. Robust standard errors, clustered at the high school level, are provided in parentheses. *** p<0.01; ** p<0.05; * p<0.1.

Table 5: Persistence Regression Results – Subgroup Analysis, Specification 2

(A) All Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
F2019	-0.12*** (0.0075)			-0.11*** (0.0059)				-0.11*** (0.0054)	-0.12*** (0.0049)	-0.12*** (0.0049)	-0.13*** (0.0066)		0.11*** (0.0051)	
F2020	-0.15*** (0.0077)			-0.14*** (0.0064)				-0.13*** (0.0054)	-0.16*** (0.0052)	-0.16*** (0.0052)	-0.16*** (0.0068)		-0.13*** (0.0057)	
F2021	-0.037*** (0.0072)			-0.057*** (0.006)				-0.035*** (0.0053)	-0.065*** (0.0052)	-0.075*** (0.0052)	-0.083*** (0.0071)		-0.0068 (0.0049)	
F2019 Interaction	-0.0016 (0.009)	-0.027** (0.011)	-0.039** (0.019)	-0.047*** (0.013)	0.064 (0.059)	0.16 (0.17)	-0.047** (0.02)	-0.011 (0.01)	-0.050** (0.024)	-0.064 (0.044)	0.0055 (0.0097)	-0.026 (0.018)	0.022* (0.013)	0.015* (0.0085)
F2020 Interaction	-0.023** (0.009)	-0.073*** (0.011)	-0.060*** (0.02)	-0.11*** (0.015)	0.33*** (0.045)	-0.14 (0.16)	-0.052** (0.022)	-0.043*** (0.011)	-0.059** (0.026)	-0.043 (0.053)	-0.0065 (0.01)	-0.013 (0.021)	0.0014 (0.014)	-0.0016 (0.0094)
F2021 Interaction	-0.064*** (0.0086)	-0.069*** (0.013)	0.0079 (0.016)	-0.079*** (0.015)	0.48*** (0.055)	-0.053 (0.17)	-0.071*** (0.02)	-0.036*** (0.012)	-0.18*** (0.026)	-0.051 (0.045)	0.024** (0.0098)	-0.11*** (0.021)	-0.10*** (0.014)	0.011 (0.009)
Mean	3.39			3.53				3.55	3.47	3.45	3.43		3.81	
Observations	345,748			345,748				345,748	345,748	345,748	345,748		345,748	
Adj. R-squared	0.25			0.25				0.23	0.24	0.25	0.25		0.11	

(B) 4-Year Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
F2019	-0.11*** (0.0077)			-0.10*** (0.006)				-0.10*** (0.0054)	-0.11*** (0.0054)	-0.11*** (0.0054)	-0.11*** (0.0076)		-0.12*** (0.0053)	
F2020	-0.12*** (0.0089)			-0.12*** (0.0064)				-0.12*** (0.006)	-0.13*** (0.0059)	-0.13*** (0.0058)	-0.12*** (0.0074)		-0.14*** (0.0057)	
F2021	0.038*** (0.0071)			0.0026 (0.0059)				0.013** (0.0051)	0.0048 (0.0051)	0.0017 (0.0051)	0.01 (0.0067)		-0.0022 (0.0049)	
F2019 Interaction	0.0061 (0.0086)	-0.0083 (0.012)	-0.031* (0.018)	-0.095*** (0.018)	0.15 (0.098)	0.077 (0.22)	-0.026 (0.021)	0.00081 (0.013)	0.039 (0.038)	-0.07 (0.085)	-0.0078 (0.011)	-0.011 (0.049)	0.083*** (0.016)	0.014 (0.0097)
F2020 Interaction	-0.018* (0.0097)	-0.052*** (0.014)	-0.064*** (0.018)	-0.15*** (0.018)	0.73*** (0.067)	-0.085 (0.18)	0.041** (0.02)	-0.02 (0.015)	0.043 (0.039)	0.021 (0.084)	-0.020* (0.012)	0.016 (0.045)	0.079*** (0.019)	0.0059 (0.011)
F2021 Interaction	-0.059*** (0.0086)	-0.037*** (0.014)	0.027** (0.013)	-0.083*** (0.018)	1.04*** (0.052)	0.071 (0.19)	0.026 (0.022)	0.011 (0.014)	-0.14*** (0.039)	-0.019 (0.079)	-0.018* (0.01)	-0.12*** (0.045)	-0.014 (0.018)	0.046*** (0.0098)
Mean	3.64			3.74				3.74	3.69	3.69	3.68		3.84	
Observations	712,588			712,588				712,588	712,588	712,588	712,588		712,588	
Adj. R-squared	0.23			0.23				0.22	0.23	0.24	0.24		0.17	

(C) 2-Year Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
F2019	-0.12*** (0.012)			-0.11*** (0.012)				-0.11*** (0.01)	-0.12*** (0.0085)	-0.12*** (0.0084)	-0.14*** (0.01)		-0.078*** (0.015)	
F2020	-0.16*** (0.012)			-0.14*** (0.013)				-0.15*** (0.01)	-0.17*** (0.0091)	-0.18*** (0.0091)	-0.18*** (0.011)		-0.11*** (0.017)	
F2021	-0.12*** (0.014)			-0.13*** (0.013)				-0.13*** (0.011)	-0.16*** (0.01)	-0.17*** (0.01)	-0.19*** (0.014)		-0.091*** (0.016)	
F2019 Interaction	-0.011 (0.016)	-0.039* (0.021)	-0.023 (0.044)	0.006 (0.02)	-0.016 (0.057)	0.2 (0.25)	-0.072* (0.039)	0.008 (0.017)	-0.063** (0.03)	-0.055 (0.05)	0.030* (0.017)	-0.045* (0.024)	-0.029 (0.021)	0.00012 (0.02)
F2020 Interaction	-0.033** (0.017)	-0.098*** (0.023)	0.031 (0.048)	-0.043* (0.025)	-0.1 (0.065)	-0.21 (0.34)	-0.18*** (0.047)	-0.024 (0.019)	-0.070** (0.033)	-0.03 (0.062)	0.0069 (0.019)	-0.027 (0.027)	-0.048** (0.023)	-0.016 (0.022)
F2021 Interaction	-0.086*** (0.018)	-0.13*** (0.025)	-0.025 (0.049)	0.001 (0.024)	-0.13 (0.085)	-0.22 (0.41)	-0.20*** (0.043)	-0.0022 (0.02)	-0.11*** (0.033)	0.018 (0.053)	0.059*** (0.02)	-0.021 (0.027)	-0.067*** (0.023)	0.027 (0.021)
Mean	3.07			3.16				3.19	3.08	3.07	3.08		3.64	
Observations	712,588			712,588				712,588	712,588	712,588	712,588		712,588	
Adj. R-squared	0.14			0.13				0.13	0.14	0.14	0.15		0.046	

Notes: These tables present the estimation results for Specification 2 using the persistence outcome variable for each type of institution: any institution (Panel A), 4-year institution (Panel B), and 2-year institution (Panel C). Each regression is separated by a dashed vertical line. Estimates are reported for each year and the interactions of year and the control variable. The mean is calculated in Fall 2018 for the control variable (e.g. female). Robust standard errors, clustered at the high school level, are provided in parentheses. *** p<0.01; ** p<0.05; * p<0.1.

Table 6: Postsecondary Paths, by Initial Enrollment Cohort and Semester**(A) Share Switching from 4-Year and 2-Year Institutions to “Other”**

	After First Semester	After Second Semester	After Third Semester
F2018	3.7 ; 16.0	6.2 ; 25.7	5.7 ; 18.6
F2019	5.2 ; 15.2	10.9 ; 34.2	11.2 ; 28.9
F2020	5.8 ; 20.0	6.6 ; 25.4	5.3 ; 19.6
F2021	5.1 ; 18.5	6.6 ; 25.3	5.3 ; 17.8

(B) Share Switching Between 4-Year Institutions and 2-Year Institutions

	After First Semester	After Second Semester	After Third Semester
F2018	2.3 ; 4.0	4.0 ; 5.7	1.9 ; 2.9
F2019	2.6 ; 2.6	2.8 ; 5.3	1.4 ; 2.6
F2020	2.2 ; 2.5	2.9 ; 7.0	1.5 ; 3.2
F2021	2.3 ; 2.6	3.4 ; 6.5	1.6 ; 3.5

(C) Share Switching from “Other” to 4-Year and 2-Year Institutions

	After First Semester	After Second Semester	After Third Semester
F2018	2.1 ; 7.3	2.5 ; 7.5	1.4 ; 5.5
F2019	1.7 ; 7.2	3.5 ; 5.1	1.4 ; 4.6
F2020	1.6 ; 6.0	3.3 ; 6.5	1.4 ; 4.6
F2021	1.9 ; 6.0	3.4 ; 6.4	1.4 ; 4.9

Notes: These tables present the share of students that transition between different enrollment states. In Panel A, the left number is the share switching from 4-year institutions to “Other” out of those enrolled in 4-year institutions the previous semester; the right number is the share switching from 2-year institutions to “Other” out of those enrolled in 2-year institutions the previous semester. In Panel B, the left number is the share switching from 4-year institutions to 2-year institutions out of those enrolled in 4-year institutions the previous semester; the right number is the share switching from 2-year institutions to 4-year institutions out of those enrolled in 2-year institutions the previous semester. In Panel C, the left number is the share of students switching from “Other” to 4-year institutions out of those in “Other” in the previous semester; the right number is the share of students switching from “Other” to 2-year institutions out of those in “Other” in the previous semester.

Figures

Figure 1: Share of High School Graduates Enrolled in Any Institution, by Graduation Year and Time from Graduation

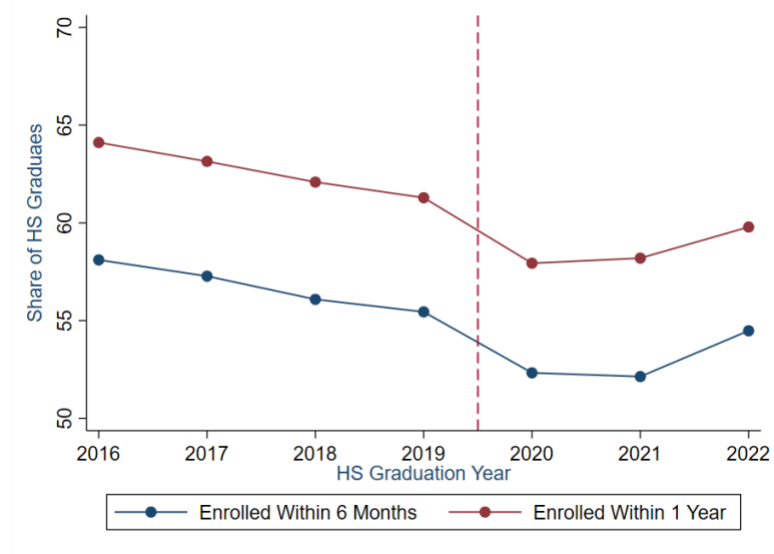
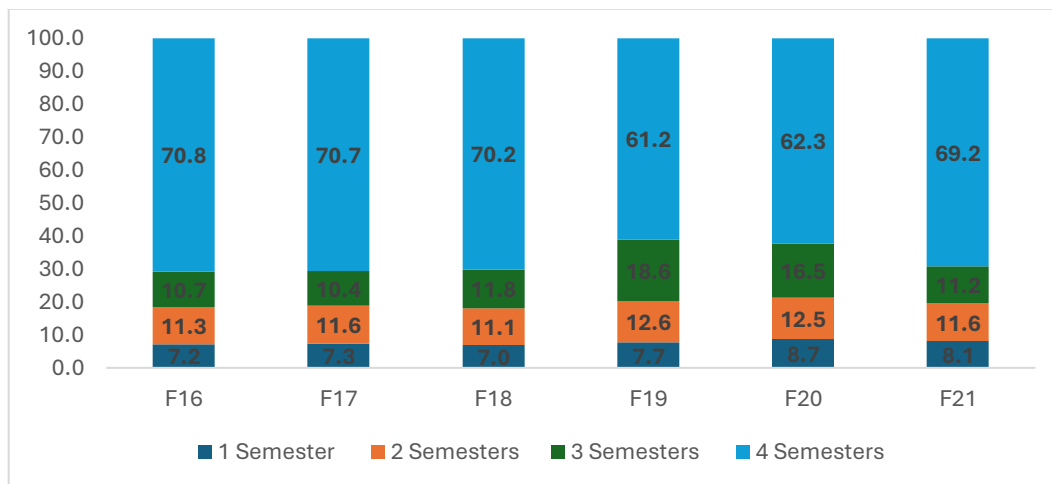


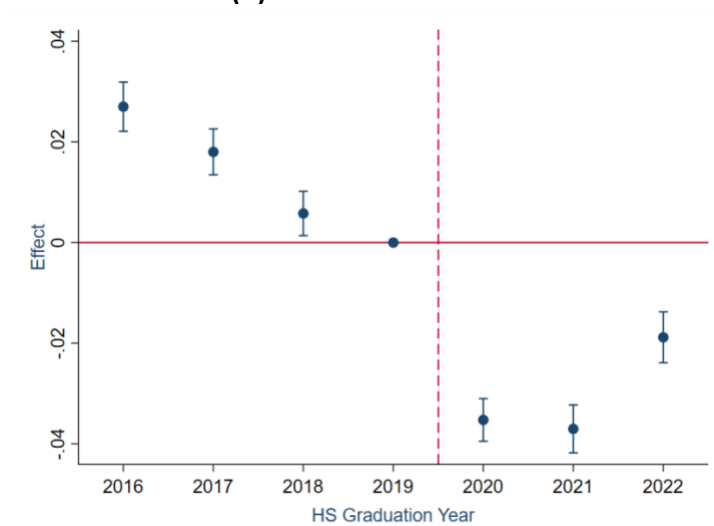
Figure 2: Numbers of Semesters Persisted in Postsecondary Institutions, by Initial Enrollment Semester



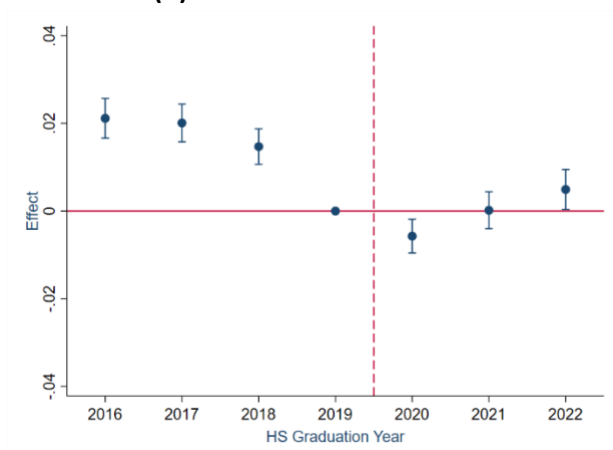
Notes: This figure details the share of postsecondary students that persisted either 1,2,3, or 4 semesters following, and including, their initial enrollment semester, by initial enrollment semester. We limit our sample to students enrolled in the fall semester. We exclude summer semesters. We include enrollment in any postsecondary institution, not just students' initial enrollment institution.

Figure 3: Enrollment Regression Results – Without Interactions, Specification 1

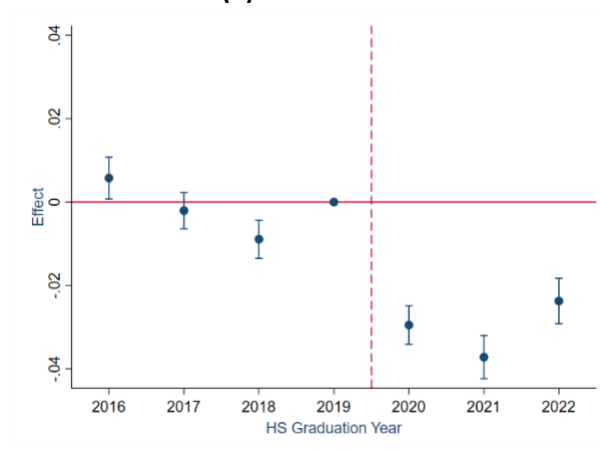
(A) All Institutions



(B) 4-Year Institutions



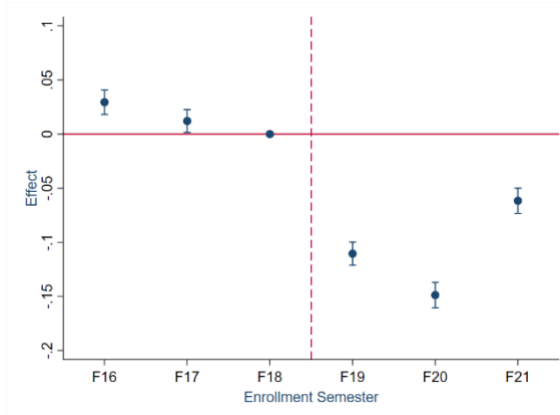
(C) 2-Year Institutions



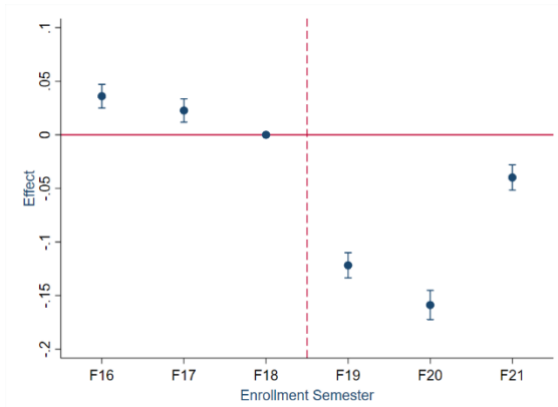
Notes: These figures present the estimation results for Specification 1 using one of the three enrollment dummy outcome variables: any institution (Subfigure A), 4-year institution (Subfigure B), and 2-year institution (Subfigure C). The regressions do not include Interactions between year and the control variables. Estimates are reported for each year, with 2019 serving as the reference year. The estimates are detailed in Table 2, columns (2)-(4). Robust standard errors are clustered at the high school level.

Figure 4: Persistence Regression Results – Without Interactions, Specification 1

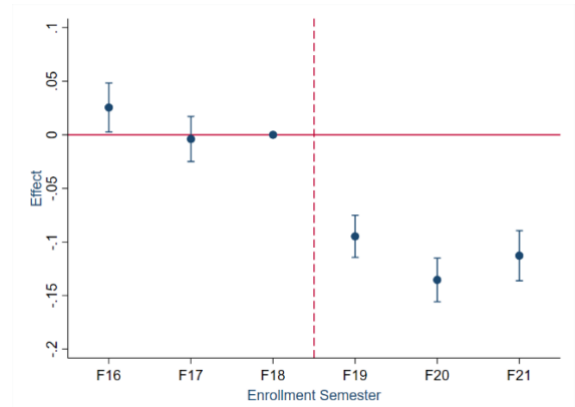
(A) All Institutions



(B) 4-Year Institutions

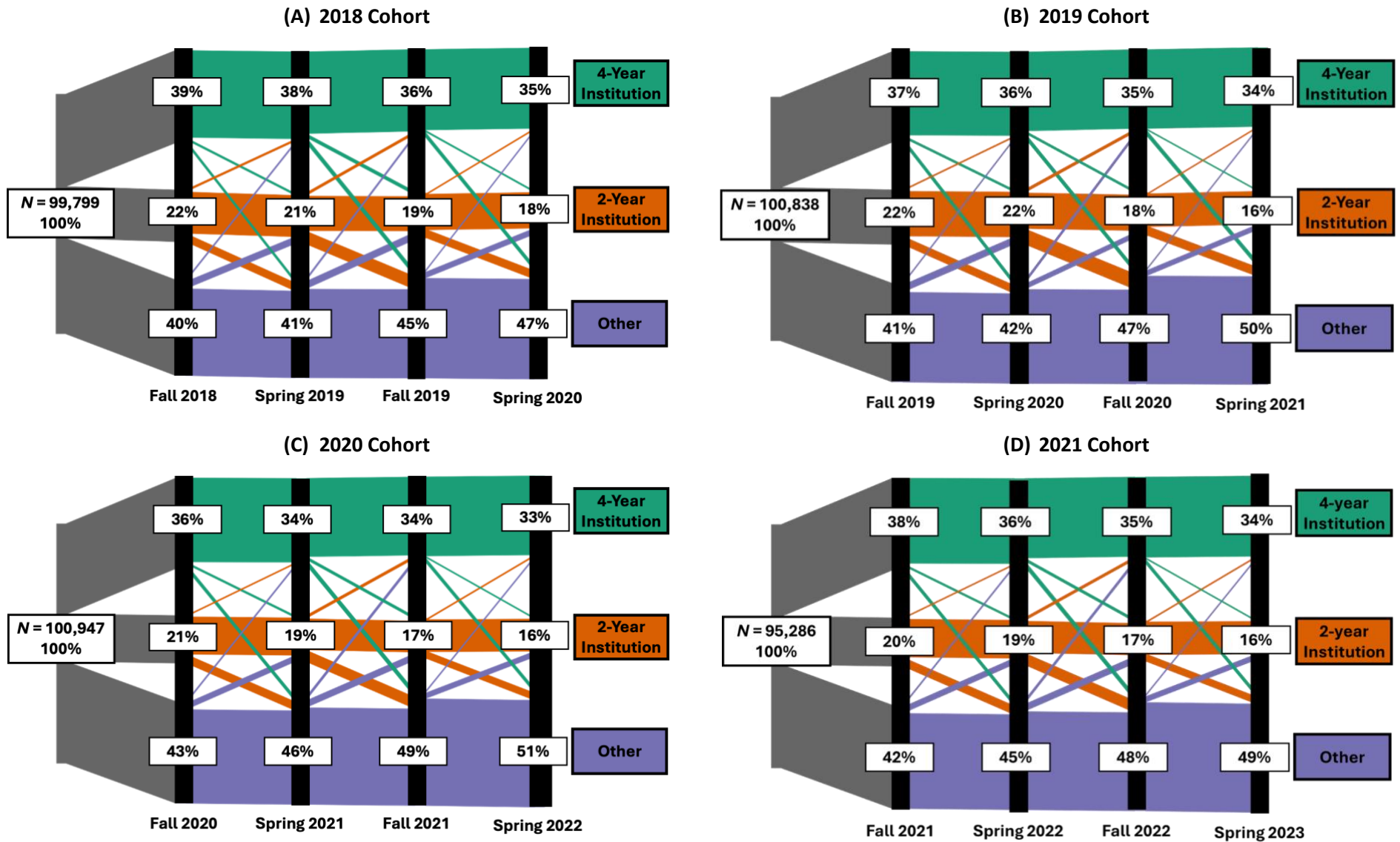


(C) 2-Year Institutions



Notes: These figures present the estimation results for Specification 1 using the persistence outcome variable, separately for each type of institution: any institution (Subfigure A), 4-year institution (Subfigure B), and 2-year institution (Subfigure C). The regressions do not include Interactions between year and the control variables. Estimates are reported for each year, with 2019 serving as the reference year. The estimates are detailed in Table 4, columns (2)-(4). Robust standard errors are clustered at the high school level.

Figure 5: Postsecondary Paths (Sankey Diagrams)



Notes: These figures track the flow of high school graduates across three different enrollment states –4-year institutions, 2-year-institutions, and “Other” (not enrolled in any postsecondary institution) – over the four semesters following high school graduation. Summer semesters are excluded. Each figure represents a different high school graduating

cohort. The shares are the proportion of each cohort that in each state at each time point. The lines connecting the different states represent student transitions, with line thickness indicating the magnitude of these movements.

Appendix

Tables

Table A1: Share of High School Graduates Enrolled in Any Institution, by Months from High School Graduation

		High School Graduates	Enrolled Within 6 Months	Enrolled Within One Year	Share Enrolled Within 6 Months	Share Enrolled Within One Year
High School Graduation Year	2015	95,515	54,606	61,087	57.17	63.96
	2016	99,719	57,809	63,792	57.971	63.97
	2017	101,787	58,216	64,187	57.19	63.06
	2018	106,428	59,581	65,963	55.98	61.98
	2019	107,868	59,732	66,026	55.38	61.21
	2020	108,551	56,628	62,697	52.17	57.76
	2021	100,740	53,960	59,344	53.56	58.91

Notes: The number of HS graduates includes only those who were matched between the NC DPI and NSC files, thus it slightly underestimates the true number of HS graduates .

Table A2: Enrollment Regression Results – Subgroup Analysis, Specification 1

(A) All Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
2016 Interaction	-0.011*** (0.0037)	0.018*** (0.0051)	-0.014 (0.011)	-0.015** (0.0065)	-0.061*** (0.017)	-0.13 (0.084)	0.0044 (0.0082)	0.0049 (0.0043)	0.0086 (0.0071)	-0.019 (0.015)	-0.0067 (0.0049)	0.025*** (0.005)	0.013** (0.0052)	0.015*** (0.0046)
2017 Interaction	-0.0047 (0.004)	0.029*** (0.0045)	-0.0013 (0.011)	0.00019 (0.006)	-0.0039 (0.014)	0.02 (0.082)	0.0076 (0.0087)	0.017*** (0.0041)	0.01 (0.007)	-0.024* (0.012)	-0.0047 (0.0048)	0.023*** (0.0052)	0.013** (0.0055)	0.017*** (0.0044)
2018 Interaction	-0.005 (0.0036)	0.015*** (0.0049)	-0.016* (0.0091)	-0.0077 (0.0059)	-0.036 (0.024)	-0.12 (0.073)	-1.2E-06 (0.0078)	0.00035 (0.0043)	0.0055 (0.007)	-0.00051 (0.01)	-0.0047 (0.0046)	-0.004 (0.0049)	-0.0054 (0.0052)	0.0044 (0.0045)
2019 Interaction (omitted)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2020 Interaction	0.014*** (0.0036)	0.00059 (0.0049)	0.0062 (0.0097)	-0.013** (0.0057)	-0.036** (0.017)	-0.0086 (0.065)	-0.0088 (0.0082)	-0.011*** (0.0042)	-0.012* (0.0064)	-0.0053 (0.011)	0.0055 (0.0043)	-0.033*** (0.0047)	-0.031*** (0.0053)	-0.020*** (0.0046)
2021 Interaction	-0.0029 (0.0036)	0.0015 (0.005)	0.00044 (0.0098)	-0.011* (0.0061)	-0.028* (0.016)	-0.16** (0.068)	-0.018** (0.0089)	0.0048 (0.0046)	0.0053 (0.0067)	0.0038 (0.01)	0.012** (0.0048)	-0.034*** (0.0053)	-0.029*** (0.0058)	-0.012** (0.0046)
2022 Interaction	0.0044 (0.0036)	-0.0021 (0.0052)	-0.020** (0.0093)	0.0069 (0.0066)	-0.047*** (0.015)	-0.17** (0.076)	-0.0064 (0.0089)	0.00095 (0.0048)	-0.0078 (0.0073)	-0.0044 (0.0097)	0.0092* (0.0053)	-0.021*** (0.0055)	-0.006 (0.006)	0.012** (0.0049)
Mean	0.545				0.681			0.700	0.642	0.624	0.603		0.913	
Observations	712,588				712,588			712,588	712,588	712,588	712,588		712,588	
Adj. R-squared	0.29				0.29			0.29	0.29	0.29	0.29		0.29	

(B) 4-Year Institutions

Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
2016 Interaction	0.002	0.016***	-0.042***	-0.020***	-0.044***	-0.034	-0.0025	-0.00078	-0.0097**	-0.040***	-0.013***	-0.039***	-0.042***	-0.018***

	-0.0036	(0.0048)	(0.01)	(0.0051)	(0.013)	(0.074)	(0.0076)	(0.0036)	(0.0047)	(0.0082)	(0.0046)	(0.0046)	(0.0052)	(0.0058)
2017 Interaction	-0.0012	0.031***	-0.031***	-0.0057	-0.011	0.017	0.0029	0.011***	-0.00083	-0.013*	-0.0072	-0.010**	-0.0076	0.0052
	-0.0035	(0.0046)	(0.0094)	(0.005)	(0.013)	(0.07)	(0.0077)	(0.0035)	(0.0047)	(0.0069)	(0.0045)	(0.0051)	(0.0057)	(0.0058)
2018 Interaction	0.0014	0.018***	-0.033***	-0.0021	-0.025**	-0.013	0.0016	0.0077**	0.0059	-0.015**	-0.0079*	-0.014***	-0.019***	-0.0095*
	-0.0035	(0.0044)	(0.0093)	(0.0047)	(0.013)	(0.061)	(0.0072)	(0.0035)	(0.0042)	(0.0073)	(0.0041)	(0.0047)	(0.005)	(0.0056)
2019 Interaction	-	-	-	-	-	-	-	-	-	-	-	-	-	-
(omitted)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2020 Interaction	0.0070*	0.018***	0.0013	0.015***	-0.024*	0.064	0.0054	0.015***	0.0074*	0.004	-0.0031	0.017***	0.0079	0.0088
	-0.0036	(0.0042)	(0.0084)	(0.0044)	(0.013)	(0.059)	(0.0074)	(0.0033)	(0.0042)	(0.0059)	(0.004)	(0.0042)	(0.0052)	(0.0055)
2021 Interaction	0.0057	0.023***	-0.0098	0.013***	0.0035	-0.0045	-0.00021	0.021***	0.0087*	0.012*	-0.00098	0.018***	0.021***	0.016***
	-0.0035	(0.0047)	(0.01)	(0.0046)	(0.011)	(0.054)	(0.008)	(0.0035)	(0.0045)	(0.0068)	(0.0044)	(0.0047)	(0.0054)	(0.0059)
2022 Interaction	0.0090**	0.0084*	-0.019*	0.012**	0.0013	-0.0005	0.0044	0.015***	-0.0064	-0.0011	-0.010**	0.0077	0.015**	0.025***
	-0.0038	(0.005)	(0.011)	(0.0051)	(0.016)	(0.056)	(0.008)	(0.0039)	(0.0047)	(0.0073)	(0.0048)	(0.0051)	(0.0057)	(0.0061)
Mean	0.299	0.410						0.439	0.380	0.364	0.323	0.768		
Observations	712,588	712,588						712,588	712,588	712,588	712,588	712,588		
Adj. R-squared	0.38	0.38						0.38	0.38	0.38	0.38	0.38		

(C) 2-Year Institutions

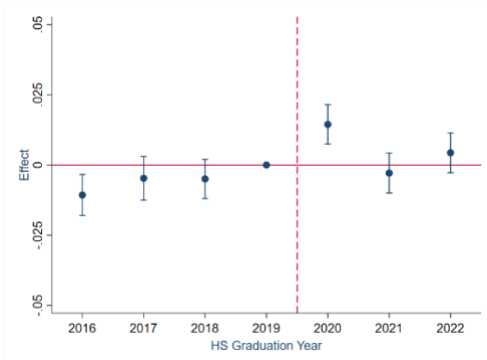
Regression	(1)	(2)						(3)	(4)	(5)	(6)	(7)		
	Female	Black	Asian	Hispanic	Am. In.	Pac. Isl.	Multi-R	Econ Dis.	Disabled	English L.	Urban	GPA Q1	GPA Q2	GPA Q3
2016 Interaction	-0.013***	0.002	0.028***	0.0049	-0.017	-0.098	0.0069	0.0055	0.018**	0.021	0.0061	0.063***	0.054***	0.032***
	(0.004)	(0.0051)	(0.011)	(0.0072)	(0.02)	(0.083)	(0.0083)	(0.0044)	(0.0072)	(0.016)	(0.0051)	(0.0056)	(0.0057)	(0.0056)
2017 Interaction	-0.0035	-0.0026	0.030***	0.0059	0.0069	0.0027	0.0046	0.0062	0.011	-0.011	0.0025	0.034***	0.021***	0.012**
	(0.004)	(0.0045)	(0.01)	(0.0064)	(0.014)	(0.092)	(0.0082)	(0.0042)	(0.0074)	(0.012)	(0.0044)	(0.0058)	(0.0055)	(0.0052)
2018 Interaction	-0.0064	-0.0028	0.017*	-0.0054	-0.011	-0.1	-0.0016	-0.0074*	-0.00039	0.014	0.0033	0.010*	0.013**	0.014***
	(0.0039)	(0.0047)	(0.0094)	(0.0066)	(0.029)	(0.067)	(0.0079)	(0.0041)	(0.0069)	(0.011)	(0.0046)	(0.0054)	(0.0052)	(0.0053)
2019 Interaction	-	-	-	-	-	-	-	-	-	-	-	-	-	-
(omitted)	-	-	-	-	-	-	-	-	-	-	-	-	-	-
2020 Interaction	0.0074*	-0.019***	0.005	-0.028***	-0.013	-0.072	-0.014*	-0.026***	-0.019***	-0.0093	0.0086*	-0.050***	-0.039***	-0.029***
	(0.0039)	(0.0047)	(0.0091)	(0.0059)	(0.016)	(0.073)	(0.0079)	(0.0043)	(0.0063)	(0.011)	(0.0047)	(0.0051)	(0.0053)	(0.0049)
2021 Interaction	-0.0086**	-0.022***	0.01	-0.023***	-0.031**	-0.15**	-0.018**	-0.016***	-0.0034	-0.0082	0.013**	-0.053***	-0.049***	-0.027***
	(0.0038)	(0.0048)	(0.0099)	(0.0064)	(0.015)	(0.068)	(0.0083)	(0.0044)	(0.0069)	(0.011)	(0.0053)	(0.0057)	(0.0058)	(0.0052)
2022 Interaction	-0.0047	-0.010**	-8.7E-05	-0.0049	-0.049***	-0.17**	-0.011	-0.014***	-0.0014	-0.0033	0.019***	-0.028***	-0.021***	-0.013**
	(0.004)	(0.0052)	(0.01)	(0.0068)	(0.018)	(0.069)	(0.0086)	(0.0048)	(0.007)	(0.011)	(0.0057)	(0.0057)	(0.0062)	(0.0052)
Mean	0.246	0.270						0.261	0.262	0.260	0.279	0.145		
Observations	712,588	712,588						712,588	712,588	712,588	712,588	712,588		
Adj. R-squared	0.29	0.29						0.29	0.29	0.29	0.29	0.29		

Notes: These tables present the estimation results for Specification 1 using one of the enrollment dummy outcomes variables: any institution (Panel A), 4-year institution (Panel B), and 2-year institution (Panel C). Each regression is separated by a dashed vertical line. Estimates are reported for the interactions of year and the control variable. The mean is calculated in 2019 for the omitted category (e.g. male for column (1)). Robust standard errors, clustered at the high school level, are provided in parentheses.

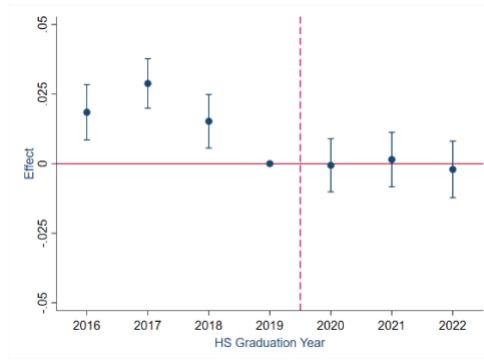
Figures

Figure A1: Enrollment Regression Results – Subgroup Analysis, All Institutions, Specification 1

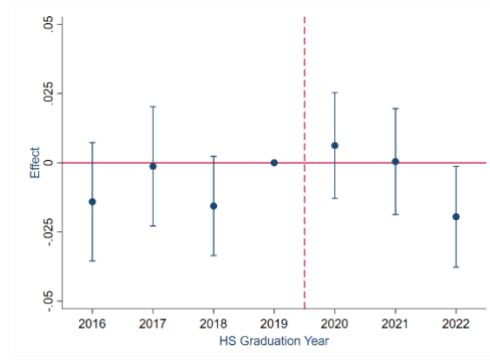
(A) Female



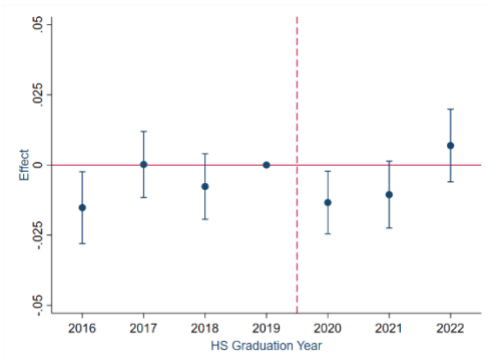
(B) Black



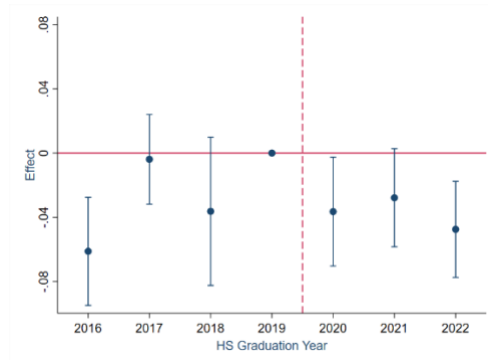
(C) Asian



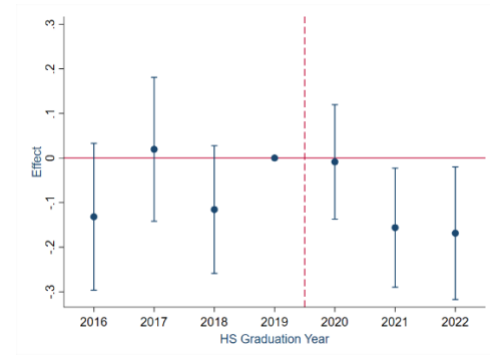
(D) Hispanic



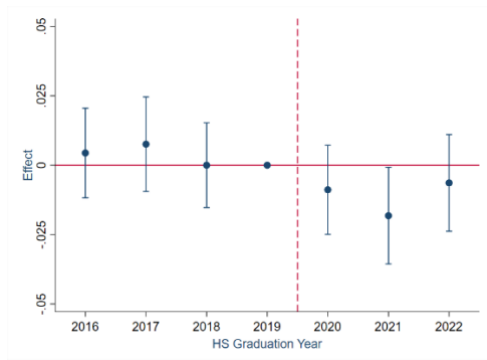
(E) American Indian



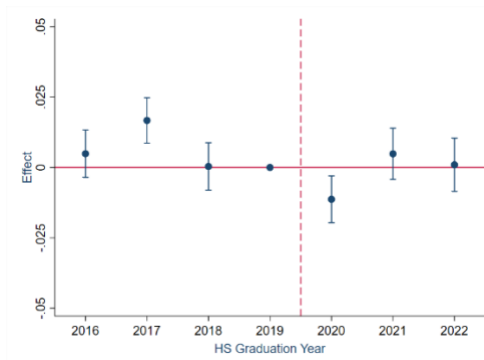
(F) Pacific Islander



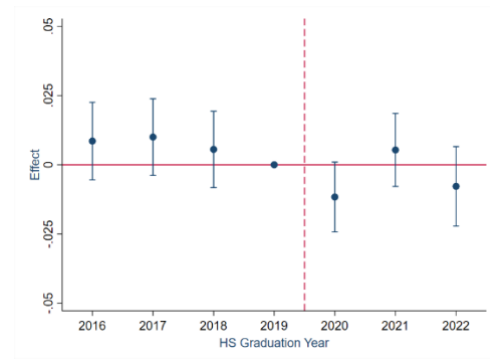
(G) Multi-Racial



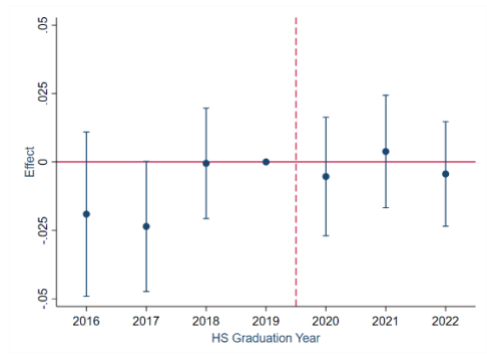
(H) Economically Disadvantaged



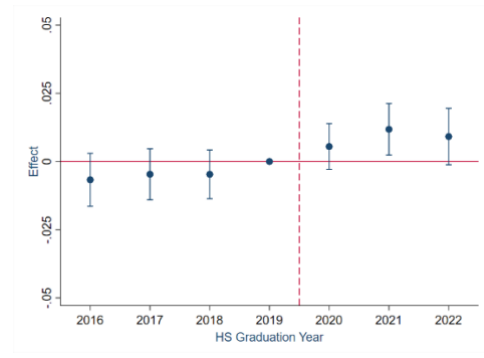
(I) Disabled



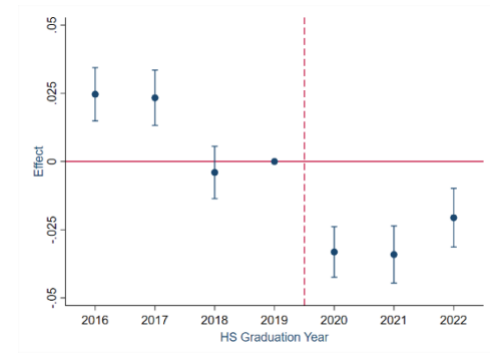
(J) English Learner



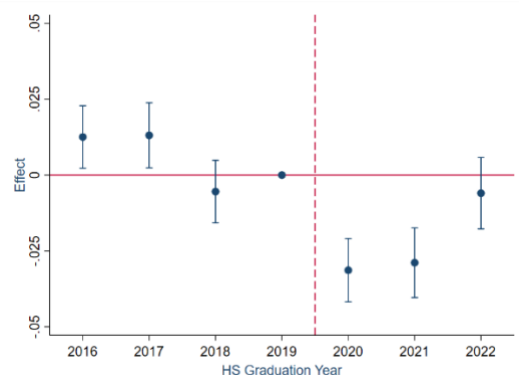
(K) Urban



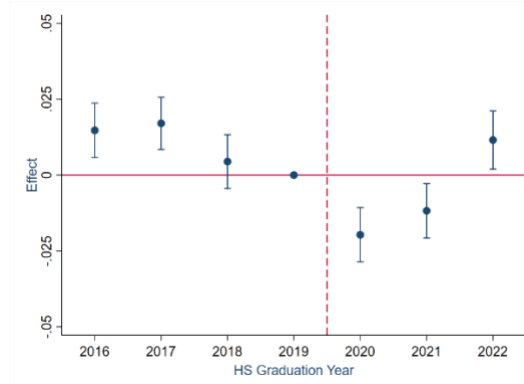
(L) GPA Quartile 1



(M) GPA Quartile 2



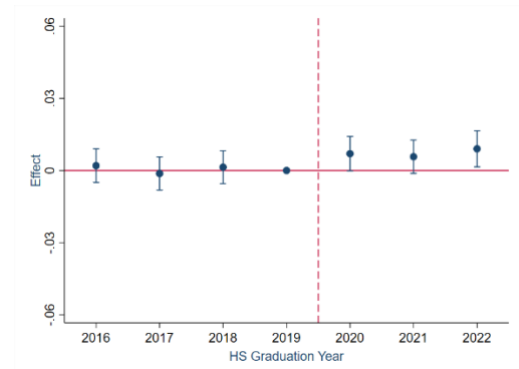
(N) GPA Quartile 3



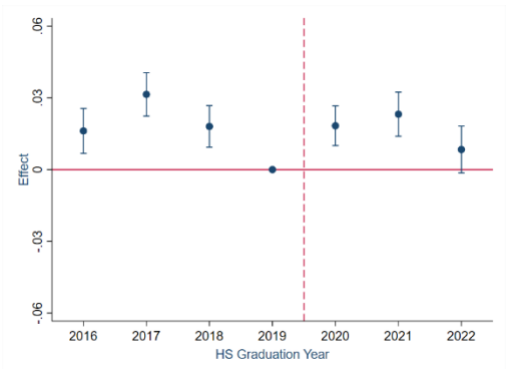
Notes: These figures present the estimation results for Specification 1 using the enrolled in any institution dummy outcome variables. Each panel reports estimates of the interactions of year and the control variable. The estimates are detailed in Pabel A of Table A2. Robust standard errors are clustered at the high school level.

Figure A2: Enrollment Regression Results – Subgroup Analysis, 4-Year Institutions, Specification 1

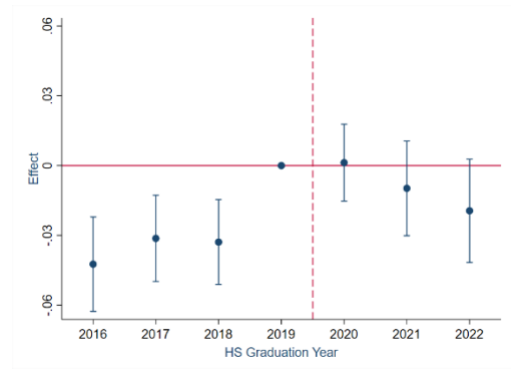
(A) Female



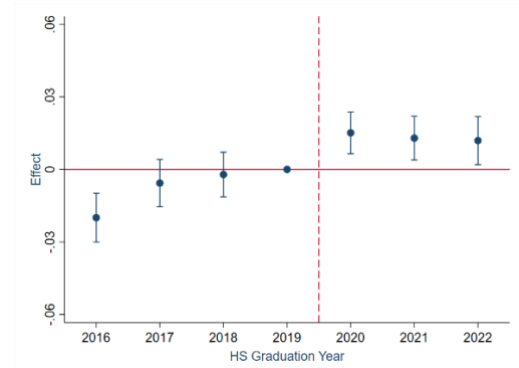
(B) Black



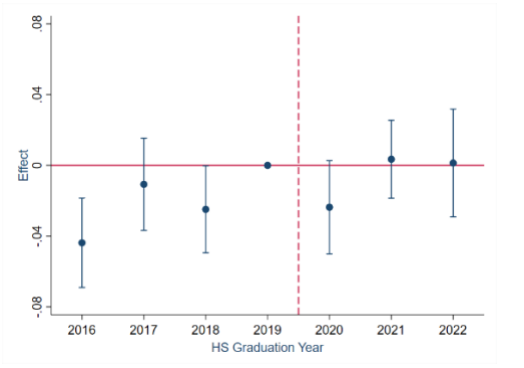
(C) Asian



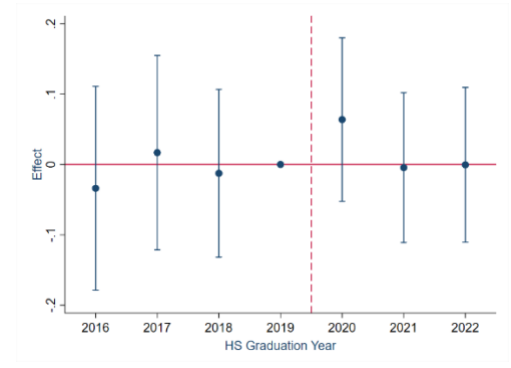
(D) Hispanic



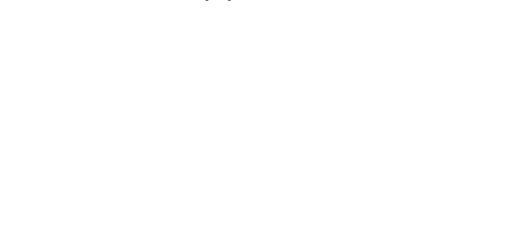
(E) American Indian



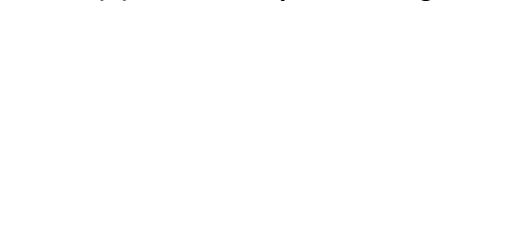
(F) Pacific Islander



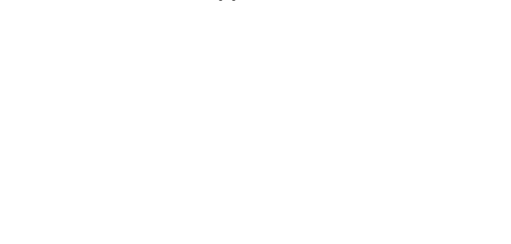
(G) Multi-Racial

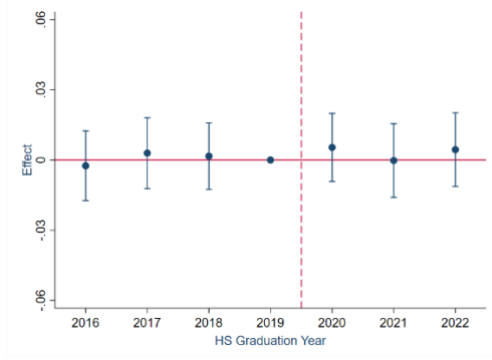


(H) Economically Disadvantaged

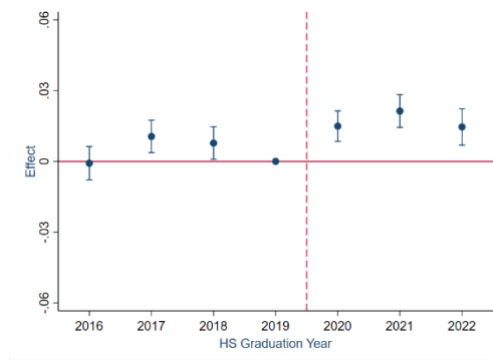


(I) Disabled

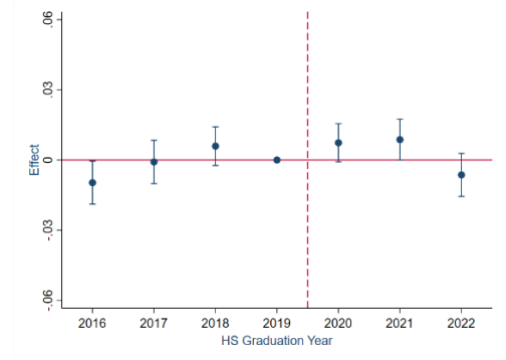




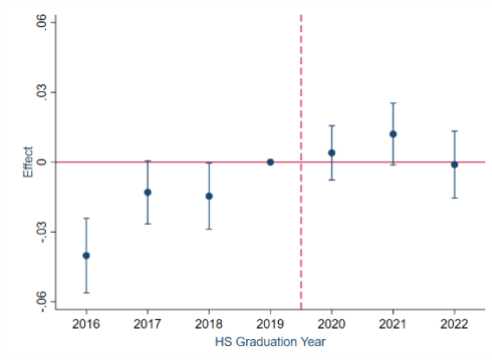
(J) English Learner



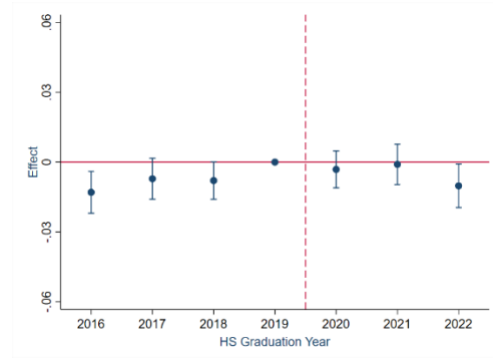
(K) Urban



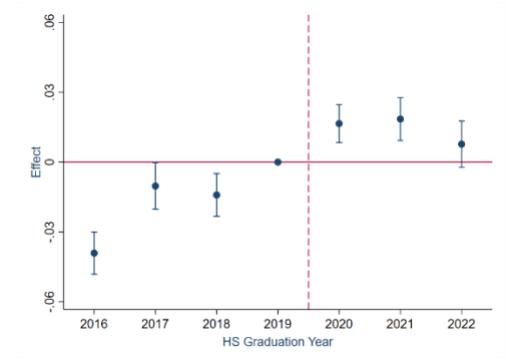
(L) GPA Quartile 1



(M) GPA Quartile 2

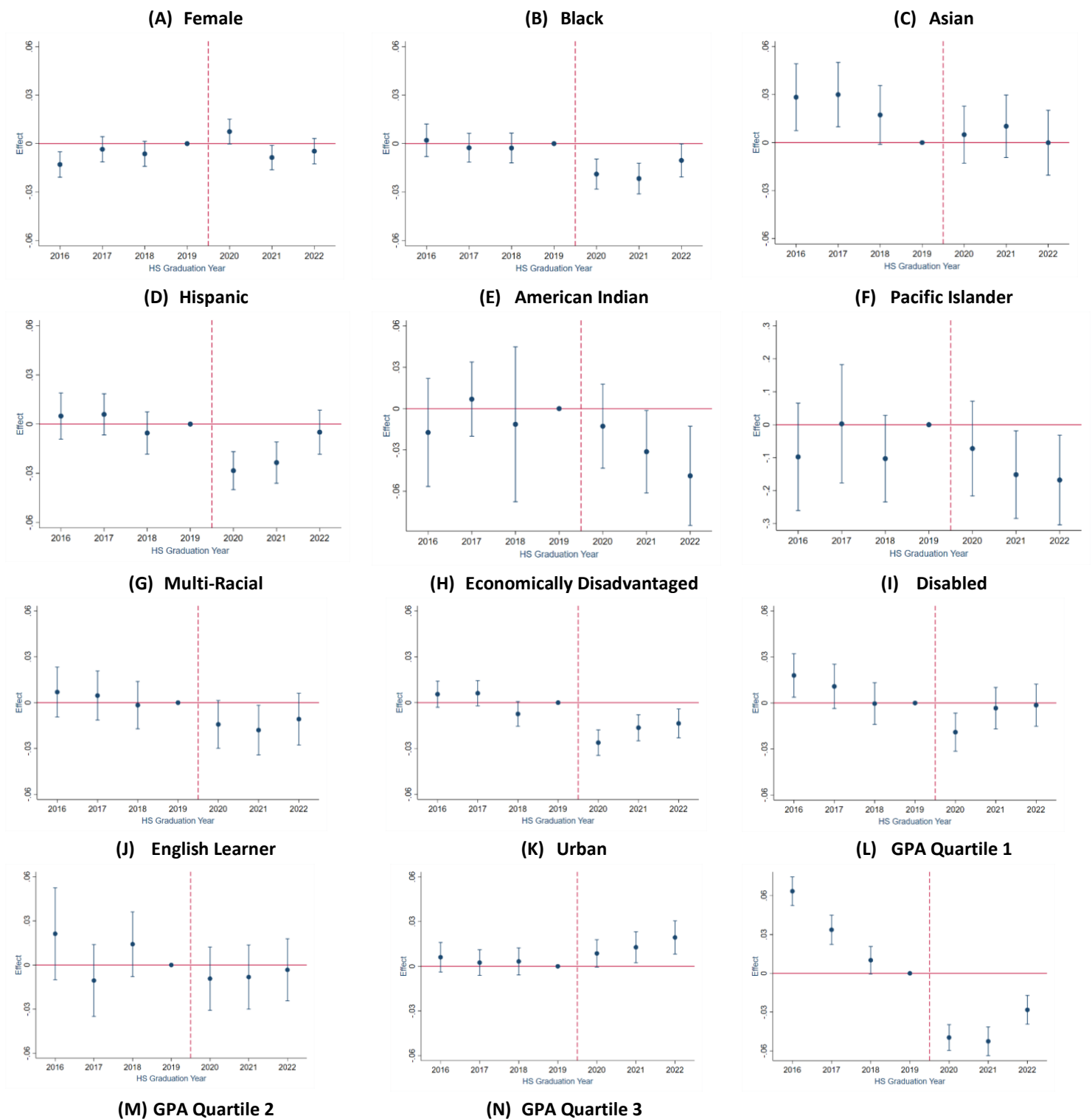


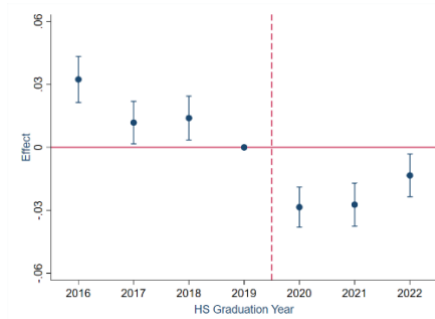
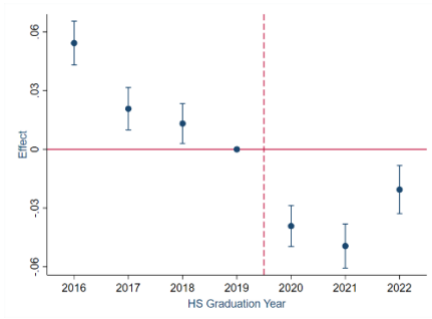
(N) GPA Quartile 3



Notes: These figures present the estimation results for Specification 1 using the enrolled in 4-year institution dummy outcome variables. Each panel reports estimates of the interactions of year and the control variable. The estimates are detailed in Pabel B of Table A2. Robust standard errors are clustered at the high school level.

Figure A3: Enrollment Regression Results – Subgroup Analysis, 2-Year Institutions, Specification 1





Notes: These figures present the estimation results for Specification 1 using the enrolled in 4-year institution dummy outcome variables. Each panel reports estimates of the interactions of year and the control variable. The estimates are detailed in Pabel C of Table A2. Robust standard errors are clustered at the high school level.